An Extraction Method for the Multipath Characteristics of Simulated Tropospheric Propagation Channels

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An Extraction Method for the Multipath Characteristics of Simulated Tropospheric Propagation Channels

Hang Zhou and Alexandre Chabory

***Abstract*—Multipath propagation, resulting from signal reflections, diffractions, scatterings, and atmospheric ducting, poses significant challenges in wireless communication, navigation, and surveillance systems. This study focuses on evaluating the multipath characteristics in tropospheric propagation channels, particularly in air-ground systems. For long-range wave propagation in low-troposphere environments, split-step methods offer both precision and computational efficiency in modeling propagation at specific frequencies. We propose a multipath extraction approach to separate dominant propagation paths and associated multipath components by means of the post-processing of frequency-domain simulation results. The multipath extraction relies on the Prony's method, which allow to assess the multipath amplitude, phase, and delay from simulation results at a few frequencies. This research examines the multipath extraction method and validates the proposed models through numerical experiments across various scenarios and parameters. Simulation results are consistent with the two-ray model over flat ground. In this case, the extracted multipath reproduces the simulated channel with a root mean square error below –50 dB at all spatial points.**

***Index Terms*—multipath extraction, split-step wavelet, Prony's method, air-ground propagation channel, urban air mobility**

I. Introduction

Urban air mobility (UAM) will provide urban, suburban, and intra-urban air transport services, using vehicles that carry a limited number of passengers [1]. As UAM is integrated into existing airspace, challenges will arise, particularly in shared airspace with other types of vehicles. At high altitudes, the airspace will be shared with conventional piloted aircraft (airplanes and helicopters). At lower altitudes, the airspace will be shared with unmanned aircraft systems (UAS), which are increasingly used for many services, such as package delivery and ground surveillance.

The use of conventional air traffic control (ATC) communication-navigation-surveillance (CNS) systems for UAM at low altitudes is questionable [2], [3]. Indeed, the maximum capacity of some systems in densely populated areas is almost reached. This is notably the case for VHF communications and for automatic dependent surveillance–broadcast (ADS-B). For these systems, it will not be possible to integrate hundreds of new vehicles. Additionally, at low altitudes, the atmosphere, terrain, and obstacles such as buildings, bridges, towers, and generally large-size structures will inevitably partially block, reflect, and scatter the signal. These perturbations will be particularly significant when the radio link involves a ground station. [4]

Therefore, it is of primary importance to develop air-ground propagation channel models that are consistent with existing and future navigation and radar systems for unmanned aerial vehicle (UAV) and UAM. Extracting delays in the multipath is important in that case. For radar systems, paths whose delays differ significantly from the direct path yield spurious echoes. For navigation systems where the time of arrival or the time-difference of arrival are used, delayed multipath impairs the positioning accuracy. Besides, UAV and UAM operate in the very low troposphere, which is the domain where the environment (atmosphere and terrain) is most likely to produce multipath.

In moderately populated areas, particularly in the context of intra-urban air transport, the system range can extend beyond several tens of kilometers. In such situations, both the terrain and the atmosphere affect the propagation channels in terms of attenuation and delay. In this case, two-dimensional methods are suitable for modeling wave propagation.

The propagation channel model can be based either on measurements or simulations. Models based on measurements require extensive measurement campaigns, using, for example, a channel sounding system [5]–[7]. The limitations of this approach are their cost and the fact that they are site-specific, i.e., the developed model will only be consistent with situations that are similar to the initial measurement environment. To assess the multipath parameters from measurements, superresolution algorithms are generally used [8]–[12]. These assume that the received signal is a sum of independent signals in the presence of white noise. This works well for channel soundings conducted with a large enough frequency band and a sufficient number of samples.

Alternatively, computational electromagnetics offer several tools to simulate propagation channels [13], [14]. If the environment is known, these tools can be applied to any site where a ground antenna is intended to be placed. However, the computational cost of these tools can be significant. Additionally, many computational electromagnetic tools work in the frequency domain, which means that propagation delays are not directly obtained. Therefore, there is a need to assess multipath from electromagnetic simulations in the frequency domain. The simulation should be performed at a few frequencies for the computation load to remain acceptable.

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In the frequency domain, ray methods can be used for predicting atmospheric propagation. Delays of multipath are then simply related to the electrical length of the ray paths. However, these methods have limitations in certain conditions, especially in atmospheric ducting. When atmospheric refraction causes rays to bend and overlap, calculation errors could occur. [15] The split-step Fourier method (SSF) is another computational electromagnetics tool that works in the frequency domain and it is often used in low tropospheric environments. NASA's Urban Air Mobility Concept of Operations [16] envisions UAM aircraft operating at cruising altitudes generally between 1,200 and 5,000 feet above ground level (AGL). These altitude ranges imply that UAM primarily operates at relatively low elevation angles when communicating with ground-based CNS systems. As a result, they are more vulnerable to multipath propagation effects caused by atmospheric ducting, surface reflections, and terrain irregularities. In contrast, for near-zenith propagation paths, the risk of multipath delays is small. SSF employs fast Fourier transforms (FFT) to speed up wave transformations between spatial and spectral domains over distances [13], [17]. To enhance the computational efficiency, the split-step wavelet (SSW) method replaces FFT with fast wavelet transform (FWT), iteratively transitioning between spatial and wavelet representations [14]. SSW reduces computational burden while maintaining similar modeling capabilities for inhomogeneous refractivity, ground boundary conditions, and irregular terrain [18], [19].

To extract multipath components, signal processing techniques can be broadly categorized into three classes. Parametric methods, such as Prony's method [20], [21], the Matrix Pencil method [22], MUSIC, and ESPRIT [23], model the received signal as a sum of complex exponentials. These methods offer high-resolution estimation of multipath parameters, including path delays and amplitudes. However, techniques like MUSIC and ESPRIT typically require a large number of frequency samples and are optimized for scenarios with additive noise. Time-domain inversion techniques aim to reconstruct impulse responses or estimate multipath parameters directly from observed signals. Representative approaches include deconvolution, sparse recovery, matching pursuit, and CLEAN-based algorithms [24], [25]. In particular, IFFT-based reconstruction provides an efficient way to convert frequency-domain data into approximate time-domain responses. While these methods are computationally efficient and easily interpretable, their resolution is fundamentally constrained by the signal bandwidth and frequency sampling density. Machine learning–based methods, including convolutional neural networks (CNNs), graph-based models, and transformers, have also been applied to extract propagation parameters from electromagnetic simulation data [26]–[28]. These data-driven approaches exhibit strong generalization capabilities across complex environments. However, they typically require large volumes of labeled training data and may lack physical interpretability. Overall, these three categories of techniques offer complementary strengths for analyzing and extracting multipath effects in electromagnetic wave propagation, each suited to different application contexts and data characteristics.

The objective of this article is to assess the multipath characteristics of the low-troposphere propagation channel. The novelty of this work lies in combining split-step methods for computing the fields at a small number of frequencies with an efficient method for extracting multipath components in terms of amplitude, phase, and delay. Several numerical experiments are conducted, and the flat-ground scenario is validated against the two-ray model.

The structure of the article is as follows: Section II outlines the simulation methodologies, detailing the split-step Fourier and split-step wavelet methods. Section III delves into the multipath extraction techniques. Section IV presents numerical simulations conducted across various configurations to evaluate the proposed multipath extraction methods.

II. Presentation of simulation methods

In this section, we first present the propagation configuration and then introduce two split-step simulation methods for calculating wave propagation in the low troposphere: the split-step Fourier method and the split-step wavelet method.

A. Configuration

The objective is to simulate the propagation in a vertical plane (x, z) and y is the direction of invariance. As plotted in Fig. 1, the source is located at $x < 0$, and the electromagnetic field is known at $x = 0$. For a wave mainly propagating along the $+x$ direction, a field with reduced phase variation, denoted as $u(x, z)$, is used. The propagation is computed in the region $x > 0$ and $z \geq 0$. We assume a time dependence of the form $e^{j\omega t}$ for the fields, where ω is the angular frequency. Any field can be decomposed into the transverse electric (TE) and transverse magnetic (TM) components with respect to the z-direction. For simplicity, we only present the TE component. The study of the TM component follows a similar approach.

For numerical purposes, the computational domain is discretized and finite in size. The vertical domain is bounded by $z \in [0, z_{\max})$ and the horizontal range is limited to $x \in [0, x_{\max})$. The discretization steps are Δx and Δz, and the number of points in two direction are given by $N_x = x_{\max}/\Delta x$ and $N_z = z_{\max}/\Delta z$. The following uniform grid is used for the discretization

$$\begin{aligned} x &= p_x \Delta x \quad \text{for} \quad p_x = \{0, \ldots, N_x - 1\}, \\ z &= p_z \Delta z \quad \text{for} \quad p_z = \{0, \ldots, N_z - 1\}. \end{aligned} \tag{1}$$

The top boundary is set as the truncation boundary of the domain. On that boundary, the Sommerfeld radiation condition should be satisfied. The ground boundary is modeled as either a perfect electric conductor (PEC) or an impedance boundary, with an irregular relief considered.

An inhomogeneous atmosphere is assumed, with the atmospheric characteristics represented by the refractive index n. Since n is close to 1, it is convenient to use the refractivity N, defined by $N = 10^6 \times (n - 1)$.

B. Split-step Fourier method

The Parabolic Equation (PE) approximates the wave equation along a paraxial direction and is implemented using

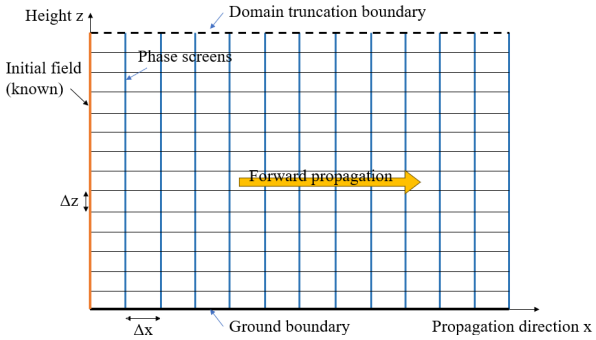


Fig. 1. Configuration of electromagnetic wave propagation simulation.

either finite differences (FD) or SSF. FD is particularly advantageous for addressing complex boundary conditions, while SSF is more computationally efficient, especially for long-range propagation scenarios. The SSF method achieves this efficiency by alternately transforming the wave field between spatial and spectral domains using FFT. In low-tropospheric wave propagation, SSF includes the modeling of atmospheric refractivity, ground characteristics, and terrain relief. To do so, phase-screen methods spatially account for refractivity variations, the discrete mixed Fourier transform accurately handles impedance ground reflections in the spectral domain, and specialized algorithms are employed to integrate terrain effects [13]. The simulation starts at x = 0 where the field is assumed to be known. This field is generally obtained either from the transmitting antenna pattern, or from an equivalent radiating aperture [13]. The propagation from x to $x + \Delta x$ is computed step by step as follows:

1) The field $u(x, z)$ undergoes transformation into the spectral domain, resulting in a vector $U(x)$ through FFT (referred to as F).
2) Free-space propagation over Δx is performed in the spectral domain by multiplying $U(x)$ by a propagator P. Thus, $U_{\text{fs}}(x+\Delta x) = PU(x)$. The wide-angle parabolic equation is applied with the range of applicable angles is approximately –45° to +45°.
3) The spatial representation of the propagated field $u_{\text{fs}}(x+\Delta x, z)$ is obtained by applying the inverse FFT (denoted by F^{-1}) on $U_{\text{fs}}(x + \Delta x)$.
4) Methods for apodization, atmosphere, and relief effects [13] are applied in the spatial domain, represented by an operator D, resulting in $u(x+\Delta x, z) = Du_{\text{fs}}(x+\Delta x, z)$.

Consequently, the propagation from x to $x+\Delta x$ is simulated step by step using

$$u(x + \Delta x, z) = DF^{-1}PFu(x, z). \tag{2}$$

These steps are iterated until the maximum computation distance x_{max} is reached.

C. Split-step wavelet method

To enhance the computational efficiency, we have introduced a wavelet-based technique called the split-step wavelet (SSW) method [14]. Subsequently, SSW has been improved in terms of memory usage and versatility by utilizing the propagation of elementary wavelets [5]. SSW employs FWT instead of FFT. The computational efficiency of FWT is significantly lower than that of FFT. SSW iteratively models electromagnetic wave propagation over increasing distances by transitioning between spatial and wavelet representations. The method proceeds as follows:

1) The field $u(x, z)$ is transformed into a sparse wavelet vector $\tilde{U}(x)$ using FWT (denoted as W) and compression (denoted as C), expressed as $\tilde{U}(x) = CWu(x, z)$.
2) Free-space propagation is performed in the wavelet domain. The coefficients vector $\tilde{U}_{\text{fs}}(x + \Delta x)$ after free-space propagation over Δx is computed using a pre-calculated free-space propagation matrix M. This matrix models wavelet-to-wavelet propagation, yielding $\tilde{U}_{\text{fs}}(x+\Delta x) = M\tilde{U}(x)$.
3) The field $u_{\text{fs}}(x + \Delta x, z)$ propagated in free space is reconstructed by the inverse FWT (W^{-1}) from the propagated wavelets: $u_{\text{fs}}(x+\Delta x, z) = W^{-1}\tilde{U}_{\text{fs}}(x+\Delta x)$.
4) Apodization, atmospheric, and relief effects, similar to those in SSF, are then applied in the spatial domain using the same operator D.

In SSW, the propagation from x to $x + \Delta x$ is simulated step by step using

$$u(x + \Delta x, z) = DW^{-1}MCWu(x, z), \tag{3}$$

and these steps are iterated until x_{max}.

Previous studies [14], [19] have shown that SSW maintains the accuracy of SSF with reduced computation time. For this reason, in this article, only simulations with SSW are considered.

III. Multipath Extraction Method

This section presents the method employed for modeling multipath signals and extracting their parameters. The accuracy of the extraction method is then analyzed. Finally, a post-processing method is proposed to group multipath components with similar delays.

A. Signal modeling

The received signal in a complex propagation environment with multiple paths can be expressed as the superposition of delayed and attenuated echoes:

$$u(f) = \sum_{p=0}^{n_{\text{p}}-1} a_p e^{-j2\pi f \tau_p}, \tag{4}$$

where f represents the electromagnetic wave frequency, n_{p} is the total number of propagation paths, and a_p and τ_p denote the complex amplitude and delay of the p-th path, respectively. This equation represents a wideband channel model, as indicated in [8]. In this study, no noise is considered, as the aim is to efficiently extract the multipath components in terms of amplitude, phase, and delay from simulation results using only a few frequency samples. The numerical errors inherent in the simulation cannot be appropriately modeled as additional Gaussian noise.

A limitation of all parametric multipath estimation methods (Prony's method, ESPRIT, MUSIC, etc.) is that the number of propagation paths n_p must be specified in advance. However, this value is generally not known a priori and cannot be reliably determined without prior knowledge of the channel structure. In typical ground-based line-of-sight propagation scenarios, the number of dominant multipath components is generally low due to the limited number of significant reflectors near the ground. Based on prior studies, we expect that a small number of paths (usually fewer than 5) are sufficient to capture the essential multipath structure.

The propagation simulation methods such as SSF and SSW are utilized to compute $u(f_n)$ at multiple regularly spaced frequencies $f_n = f + n\delta f$, where $n \in \{0, \cdots, n_\mathrm{f}-1\}$ and n_f is the total number of frequency samples. For simplicity, let $u_n = u(f_n)$.

The objective is to determine the multipath parameters a_p and τ_p for $p \in \{0, \cdots, n_\mathrm{p}-1\}$ from u_n, using as few frequency samples n_f as possible. After sampling, the problem can be reformulated as

$$u_n = \sum_{p=0}^{n_\mathrm{p}-1} a_p {z_p}^n, \qquad \text{with } z_p = e^{-j2\pi\delta f \tau_p}, \tag{5}$$

where $n = 0, \cdots, n_\mathrm{f}-1$. The right-hand-side of (5) can be cast as matrix-vector product, where the matrix elements are

$$\mathbf{M}_{\tau_0,\cdots,\tau_{n_\mathrm{p}-1}} = \begin{bmatrix} 1 & \cdots & 1 \\ z_0 & \cdots & z_{n_\mathrm{p}-1} \\ \vdots & & \vdots \\ {z_0}^{n_\mathrm{f}-1} & \cdots & (z_{n_\mathrm{p}-1})^{n_\mathrm{f}-1} \end{bmatrix}, \tag{6}$$

and the vector components are

$$\mathbf{u} = \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n_\mathrm{f}-1} \end{bmatrix}, \qquad \mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n_\mathrm{p}-1} \end{bmatrix}. \tag{7}$$

Thus, (5) can be expressed in matrix form as follows

$$\mathbf{u} = \mathbf{M}_{\tau_0,\cdots,\tau_{n_\mathrm{p}-1}} \mathbf{a}, \tag{8}$$

For simplicity, the indices $\tau_0, \cdots, \tau_{n_\mathrm{p}-1}$ are omitted from $\mathbf{M}$. The matrix $\mathbf{M}$ exhibits a structure known as a Vandermonde matrix.

B. Prony's method

Prony's method [20], [21] is based on the principle that any series in the form of (5) satisfies a homogeneous difference equation. This implies the existence of coefficients $h_1, \cdots, h_{n_\mathrm{p}}$ such that

$$u_n = h_1 u_{n-1} + h_2 u_{n-2} \ + \cdots + h_{n_\mathrm{p}} u_{n-n_\mathrm{p}} \tag{9}$$

In other words, u_n can be interpreted as the output of a finite impulse response (FIR) filter. The corresponding characteristic polynomial is given by

$$P = \sum_{p=0}^{n_\mathrm{p}} h_p z^{p-n_\mathrm{p}}, \qquad \text{with} \quad h_0 = -1. \tag{10}$$

This polynomial is related to the Z-transform of the filter. The roots of this polynomial are z_p. From (4), the multipath delays derive from the phases of these roots.

For $n_\mathrm{f} \geq 2n_\mathrm{p}$, the following steps estimate τ_p and a_p from u_n:

- Calculate h_p:
  Using (9), h_p can be obtained by solving the linear system

$$\begin{bmatrix} u_{n_\mathrm{p}-1} & \cdots & u_0 \\ u_{n_\mathrm{p}} & \cdots & u_1 \\ \vdots & & \vdots \\ u_{n_\mathrm{f}-2} & \cdots & u_{n_\mathrm{f}-n_\mathrm{p}-1} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_{n_\mathrm{p}} \end{bmatrix} = \begin{bmatrix} u_{n_\mathrm{p}} \\ \vdots \\ u_{n_\mathrm{f}-1} \end{bmatrix}. \tag{11}$$

  If $n_\mathrm{f} = 2n_\mathrm{p}$, the system is square, and the solution fits the data exactly. For $n_\mathrm{f} > 2n_\mathrm{p}$, the system is overdetermined and solved using the least squares method.
- Compute the delays τ_p:
  The roots z_p of the polynomial (10) are calculated, from which the delays are obtained as

$$\tau_p = -\frac{\angle z_p}{2\pi\delta f} \tag{12}$$

- Solve for $\mathbf{a}$:
  Using the values of z_p in $\mathbf{M}$, the linear system (8) is solved to determine $\mathbf{a}$. Here, the linear system is of size $n_f \times n_p$. It is always overdetermined since (11) imposes $n_f \geq 2n_p$.

For the special case of a single propagation path ($n_\mathrm{p} = 1$) with two frequency samples, the polynomial reduces to first-order, with the root

$$z_0 = \frac{u_1}{u_0} \qquad \Rightarrow \qquad \tau_0 = -\frac{\angle(u_1/u_0)}{2\pi\delta f}. \tag{13}$$

Since the group delay of a signal is defined by -$d\angle u/d\omega$, the Prony's method provides in this case the finite-difference approximation of the group delay. The phase ambiguity conditions are further discussed in [29].

C. Accuracy

Because of the possible phase ambiguity, the estimated phase of z_p in (12) will be $2\pi\delta f\tau_p + 2l\pi$, where l is an integer. This implies that τ_p can be directly extracted, i.e., without phase ambiguity from $\angle z_p$ if the condition $2\pi\delta f\tau_p \leq 2\pi$ is satisfied. Hence, the following criteria must be respected

$$\delta f \leq 1/\tau_p. \tag{14}$$

However, a small value of δf implies that the phase in z_p will also be very small, which may be impaired by the simulation's accuracy. Additionally, for small values of δf, the linear system that must be solved in Prony's method will be very poorly conditioned since successive values of u_n will be very similar.

D. Reduction of the number of multipath

The application of Prony's method may yield results containing very weak multipath components or multiple paths with nearly identical delays. To enhance the interpretability

of the simulation results, we propose a post-processing step to simplify the multipath representation.

First, multipath components with amplitudes below a predefined threshold a_{thres} are suppressed. The modified amplitudes are given by

$$a_p^{\text{thres}} = \begin{cases} a_p & \text{if } |a_p| > a_{\text{thres}} \\ 0 & \text{if } |a_p| \leq a_{\text{thres}}. \end{cases} \tag{15}$$

Due to various uncertainties (e.g., atmospheric refraction index, terrain relief, ground composition) and simulation approximations, the channel response is only approximately modeled. Therefore, in numerical experiments, the threshold below which paths are removed is set to -60 dB.

Second, multipath components with delays differing by less than a predefined threshold τ_{min} are grouped into a single equivalent path. Consider two multipath components, indexed as p and p', with delays τ_p and $\tau_{p'}$, respectively. If their delay difference satisfies $|\tau_p - \tau_{p'}| < \tau_{\text{min}}$, the components are merged into a single equivalent path with the following parameters

$$a^{\text{clustered}} = a_p + a_{p'}, \qquad \tau^{\text{clustered}} = \frac{|a_p|^2\tau_p + |a_{p'}|^2\tau_{p'}}{|a_p|^2 + |a_{p'}|^2}. \tag{16}$$

The choice of τ_{min} depends on the parameters of the system under study, notably its bandwidth. Moreover, for radar and navigation systems, the delay resolution typically corresponds to a path length of at least 1 m. Therefore, τ_{min} is set to $1/c_0$, where c_0 denotes the speed of light in vacuum. [30], [31]

E. Complexity analysis

The total computational complexity of the SSW method combined with the multipath extraction method, denoted as $\mathcal{C}_{\text{TC}}$, can be expressed as

$$\mathcal{C}_{\text{TC}} = n_f \mathcal{C}_{\text{SSW}} + n_x n_z \mathcal{C}_{\text{ext}}, \tag{17}$$

where

- $\mathcal{C}_{\text{SSW}}$ denotes the computational cost of the SSW method. As detailed in [32], the complexity of the fast wavelet transform and the associated compression operations is approximately $\mathcal{O}(N_x N_z)$.
- $\mathcal{C}_{\text{ext}}$ represents the complexity of the multipath extraction at a single point. This includes solving (11), computing the roots of a polynomial of degree n_p, and solving (8), a linear system of size n_f.

In the numerical experiments, only a limited number of frequencies n_f, and hence a small number of propagation paths n_p, are extracted, the overall computational complexity is generally dominated by $n_f \mathcal{C}_{\text{SSW}}$.

IV. Numerical experiments

This section explores numerical simulations using various configurations of terrain, atmospheric conditions, and simulation parameters. Prony's method described in subsection III-B and the post-processing method introduced in subsection III-D are employed to efficiently extract multipath components from the simulation results using SSW.

A. Wave propagation over a planar PEC ground

The initial simulation explores the propagation over a planar perfect electric conductor (PEC) ground in a homogeneous atmosphere. A complex source point (CSP) [33] serves as the wave's source, located at $(x_{\text{s}}, y_{\text{s}}, z_{\text{s}})$ with $x_{\text{s}} \in \mathbf{C}$ such that $x_{\text{s}} = x_{w0} - jkW_0^2/2$, where k is the wavenumber and W_0 is the source's width. In free space, the wave radiation can be analytically derived using the Green's function and its expression in 2D is given by

$$u(r) = A\frac{j}{4}H_0^{(2)}(kr), \tag{18}$$

where A is a normalization constant, $H_0^{(2)}$ is the Hankel function of the second kind, and

$$r = \sqrt{(x - x_{\text{s}})^2 + (z - z_{\text{s}})^2}, \tag{19}$$

with the condition that the real part of r is positive. CSP represents an antenna whose radiation pattern is approximately Gaussian. This could correspond to a Potter or corrugated horn antenna. Moreover, several methods exist to expand any radiation into a linear combination of Gaussian waveforms. Therefore, by extension, the method is applicable to any electromagnetic waves, including those characterized by an arbitrary CNS antenna radiation pattern.

In the presence of a PEC planar ground, an image CSP is introduced. The exact solution is obtained by superposing the fields radiated from the original CSP and the field from the image CSP, the latter of which is multiplied by -1. Two propagation paths can be defined, one corresponding to each CSP. The parameters of the wave source are listed in Table I.

TABLE I
PARAMETERS OF THE COMPLEX SOURCE POINT

Parameters	Values
EM wave frequency f	5 GHz
Polarization	TE
W_0	0.5 m
x_{w0}	-100 m
y_s	0 m
z_s	200 m

TABLE II
SIMULATION PARAMETERS USED FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION A.

Parameters	Descriptions	Values
z_{max}	Maximum altitude of the total simulation domain	1000 m
$z_{\text{max_cd}}$	Maximum altitude of the computation domain	700 m
Δ_z	Vertical discretization step	0.2 m
N_z	Number of vertical grid points	5000
x_{max}	Maximum horizontal range of the computation domain	15 km
Δ_x	Horizontal discretization step	30 m
N_x	Number of horizontal grid points	500

The simulation parameters are detailed in Table II. In SSW, an absorbing layer is placed at the upper boundary of the computation domain to prevent reflection. This layer occupies a fixed 30% of the domain. In Table II, $z_{\text{max_cd}}$ denotes

the maximum altitude of the effective computational domain, excluding the absorbing layer. The compression threshold in SSW is set to 10^{-8}, which corresponds to a maximum compression error of -100 dB, as described in [14], [34].

Prony's method is used to extract multipath components with $n_{\mathrm{p}} = 2$. To capture 2 distinct multipaths, it is necessary to propagate at least $n_{\mathrm{f}} = 4$ different frequencies, uniformly spaced by δf. Here, we always set $n_f = 2n_p$ in the simulations. According to (14), δf is set to 0.5 MHz, so that multipath with delays up to 600 m can be extracted without ambiguity. This results in a sequence of frequencies: 5.0000, 5.0005, 5.0010, and 5.0015 GHz. In the post-processing, $a_{\mathrm{thres}} = 10^{-3}$ and $\tau_{\mathrm{min}} c_0 = 1$ m are set, as introduced in subsection III-D.

The normalized propagated field is illustrated in Fig. 2. The upper zone above the dashed line corresponds to the apodization/absorbing layer. In this simulation, this zone is shown for illustration purposes only and is omitted in the results of the other simulations for simplicity. The direct and reflected waves superimpose to form interference fringes, each with different attenuation and delay. After applying the multipath extraction method, the first and second extracted path components are presented in Fig. 3. The left subplots display the group delays of the reduced field, while the right subplots depict the amplitudes corresponding to these components. To simplify interpretations of the simulations results, delays are characterized as the distance τc_0. The top row highlights the first path component, which is the dominant one. It corresponds to direct propagation, representing the wave radiated directly from the source. The second row shows the reflected wave. The grazing reflection is categorized as part of the direct propagation path rather than being grouped with the reflected waves in the second figure. This is because very near the ground and at grazing incidence, the direct and the reflected waves travel almost the same distance, that is to say, the rays are almost of the same length. Thus, the algorithm cannot separate the direct field from the ground reflection near the ground. The delays and amplitude of the reflected wave obtained using SSW, followed by the extraction method, are compared with the two-ray model, which includes both line-of-sight and ground reflection. The results for the final vertical (at the maximum range 15 km) are presented in Fig. 4a and Fig. 4b. The extraction results align with the theoretical analysis.

To validate the multipath extraction results, we evaluated the cumulative error between the reconstructed fields (obtained using (4)) and the simulated fields obtained with SSW at n_{f} frequencies. This error is quantified using the root mean square (RMS) error in dB

$$\epsilon_{\mathrm{RMS}} = 10 \log \frac{\sum_{n=0}^{n_{\mathrm{f}}-1} \left| u_n - \sum_{p=0}^{n_{\mathrm{p}}-1} a_p e^{-2j\pi n \delta f \tau_p} \right|^2}{\sum_{n=0}^{n_{\mathrm{f}}-1} |u_n|^2}. \tag{20}$$

The computed error values are plotted in Fig. 5, showing a difference of less than -50 dB within the paraxial range from the CSP. This low error magnitude confirms the validity of the proposed multipath extraction method.

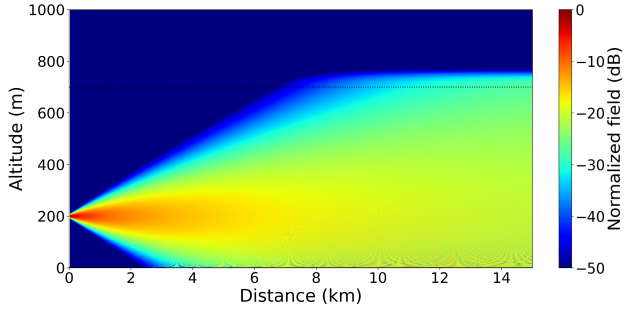


Fig. 2. Normalized electromagnetic field computed using SSW over a planar PEC ground in a homogeneous atmosphere (Source height: 200 m).

Additionally, we compare the simulation result $u(f)$ at a frequency of $f = 5000.75$ MHz with the reconstructed field obtained from using the multipath extraction method. The difference between these two fields is

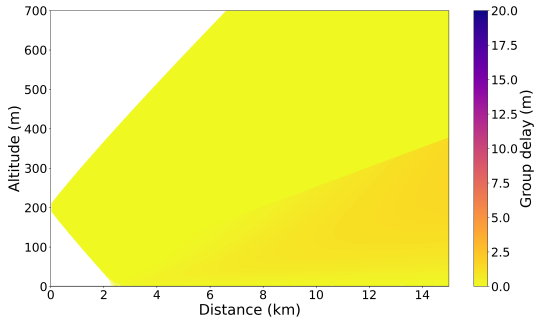
$$\epsilon'_{\mathrm{RMS}} = 10 \log \frac{\left| u(f) - \sum_{p=0}^{n_{\mathrm{p}}-1} a_p e^{-2j\pi f \tau_p} \right|^2}{|u(f)|^2}, \tag{21}$$

where the summation represents the reconstructed field composed of the multipath components. The error is plotted in Fig. 6, showing that the maximum difference between the simulation and the reconstructed field is -39.7 dB. This small difference indicates a high accuracy of the multipath extraction method.

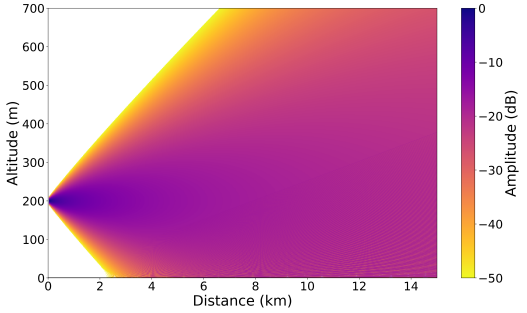
When the source height is set to 20 m, the delay between the reflected and direct paths is less than 2 m. The normalized propagated field is illustrated in Fig. 7. After applying the multipath extraction method, the first and second components are presented in Fig. 8. The amplitude of the second component is negligible (under -50 dB). In this case, only one dominant path is sufficient to characterize the channel. This shows a limitation of the method: it can effectively separate different components only when there is a sufficient delay between them.

B. Wave propagation over an irregular dielectric ground

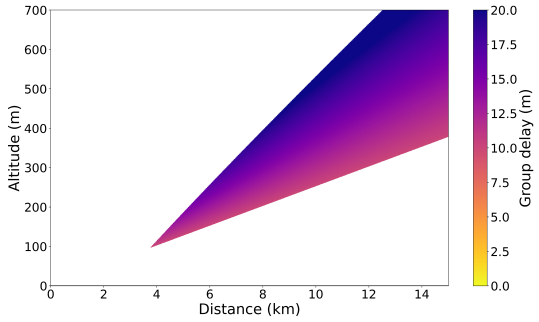
The second simulation examines the propagation characteristics of a CSP over an irregular dielectric ground. The ground profile, depicted as the black region in Fig. 9, represents a realistic terrain with a maximum height of approximately 100 m. We adopt a terrain generation method in which the terrain profile is formed by the superposition of random surfaces defined across multiple scales [35]. At each scale, random height values are sampled and interpolated, typically using linear or cosine interpolation, to create smooth transitions. The resulting multi-scale layers are then weighted and summed to produce a one-dimensional terrain that exhibits both randomness and natural continuity. The ground's relative permittivity is set to $\epsilon_{\mathrm{r}} = 15$, and its conductivity is $\sigma = 0.5$ S/m.



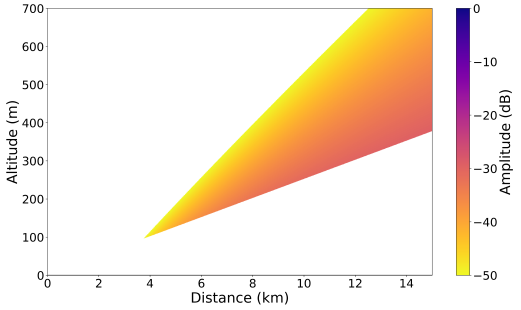
(a) Delay of the first extracted path component



(b) Amplitude of the first extracted path component

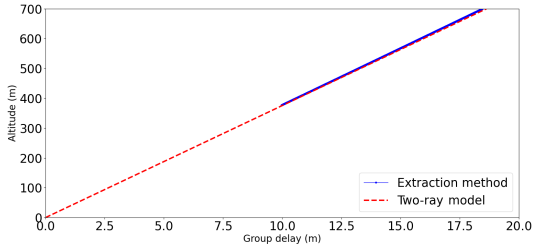


(c) Delay of the second extracted path component

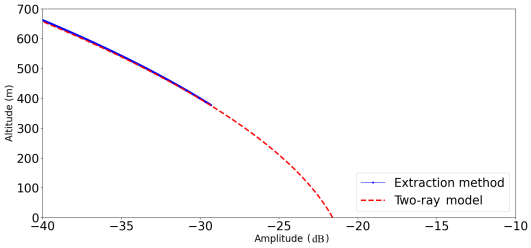


(d) Amplitude of the second extracted path component

Fig. 3. Extracted multipath components in propagation over a planar PEC ground in a homogeneous atmosphere (Source height: 200 m).



(a) Delay of the second path component: extraction method compared to analytical solution



(b) Altitude of the second path component: extraction method compared to analytical solution

Fig. 4. Comparison of the delay and altitude of the second path component between the extraction method and the analytical solution at the maximum range (15 km) over a planar PEC ground in a homogeneous atmosphere.

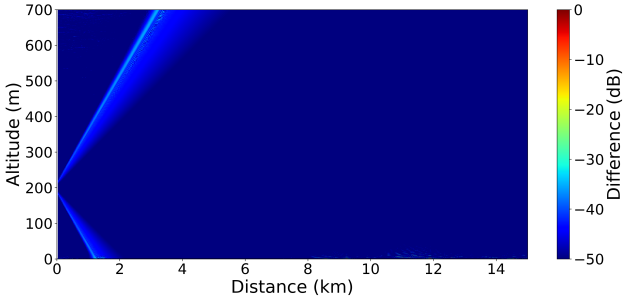


Fig. 5. Root mean square error between the reconstructed field using multipath extraction and the simulated field from SSW across n_{f} frequencies.

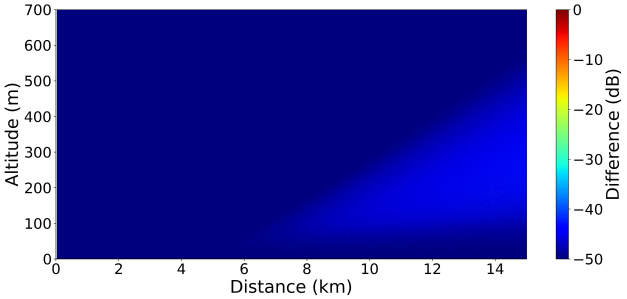


Fig. 6. Root mean square error between the reconstructed field using multipath extraction and the simulated field from SSW at the frequency 5000.75 MHz.

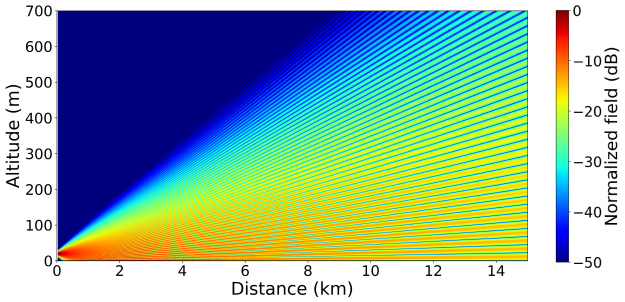


Fig. 7. Normalized electromagnetic field computed using SSW over a planar PEC ground in a homogeneous atmosphere (Source height: 20 m).

The simulation considers $n_{\mathrm{f}} = 8$ discrete frequencies, uniformly spaced by $\delta f = 5\,\mathrm{MHz}$. The normalized propagated field, calculated using SSW, is presented in Fig. 9. The extracted propagation paths, limited to $n_{\mathrm{p}} = 4$, are shown in Fig. 10. The first component in Fig. 10 corresponds to the direct wave. The next components represent multipath effects, which occur when ground reflections and diffractions generate fields that propagate in directions significantly deviating from the field emitted by the CSP.

C. Wave propagation in a complex environment

The propagation of a CSP within a complex environment is analyzed, focusing on long-range wave propagation over an irregular dielectric ground under a ducting atmosphere with accounting the Earth's curvature. The atmospheric model, illustrated in Fig. 11, is defined by a trilinear profile commonly used to represent ducting layers in the atmosphere [13]. The model is characterized by the following parameters: $z_{\mathrm{b}} = 200\,\mathrm{m}$ and $z_{\mathrm{t}} = 1000\,\mathrm{m}$, which denote the bottom and top heights of the ducting layer, respectively; and $c_1 = -0.1\,\mathrm{M\text{-}unit/m}$, $c_2 = 0.118\,\mathrm{M\text{-}unit/m}$, which represent the vertical gradients of modified refractivity within different sublayers. In this article, the Earth curvature is realized by using modified refraction index, as introduced in [13].

Details of the source configuration and simulation parameters are summarized in Table III. Simulations are performed at 8 frequencies with a frequency increment of $\delta f = 1.5\,\mathrm{MHz}$. Fig. 12 illustrates the normalized electromagnetic wave fields computed using the SSW method, while Fig. 13 presents the extracted propagation components derived from the simulations.

The electromagnetic wave propagation is significantly influenced by the refractive index gradient within the atmospheric duct. Above 200 m, waves bend upward due to the decreasing refractive index, whereas below 200 m, they are guided downward by the ducting effect. Upon interacting with the irregular terrain, reflection and diffraction phenomena produce complex multipath effects, as depicted in Fig. 12.

After applying the multipath extraction method, four path components are identified and shown in Fig. 13. The first component, represented in the top row, predominantly includes direct waves bent by the refractive index gradient within the duct. The second component mainly consists of waves reflected or diffracted by the irregular ground. The third and fourth components correspond to waves with substantial group delays, arising from interactions between direct waves and terrain-reflected waves. This classification highlights the intricate wave behavior within a ducting atmosphere.

TABLE III
SOURCE AND SIMULATION PARAMETERS FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION C.

Parameters	Values	Parameters	Values
f	2 GHz	$z_{\max}$	2000 m
Polarization	TE	$z_{\max_\mathrm{cd}}$	1400 m
W_0	5 m	Δ_z	0.2 m
x_{w0}	-100 m	N_z	10000
y_s	0 m	$x_{\max}$	250 km
z_s	200 m	Δ_x	500 m
n_{f}	8	N_x	500
δ_{f}	1.5 MHz	n_{p}	4

D. Propagation with high obstacles

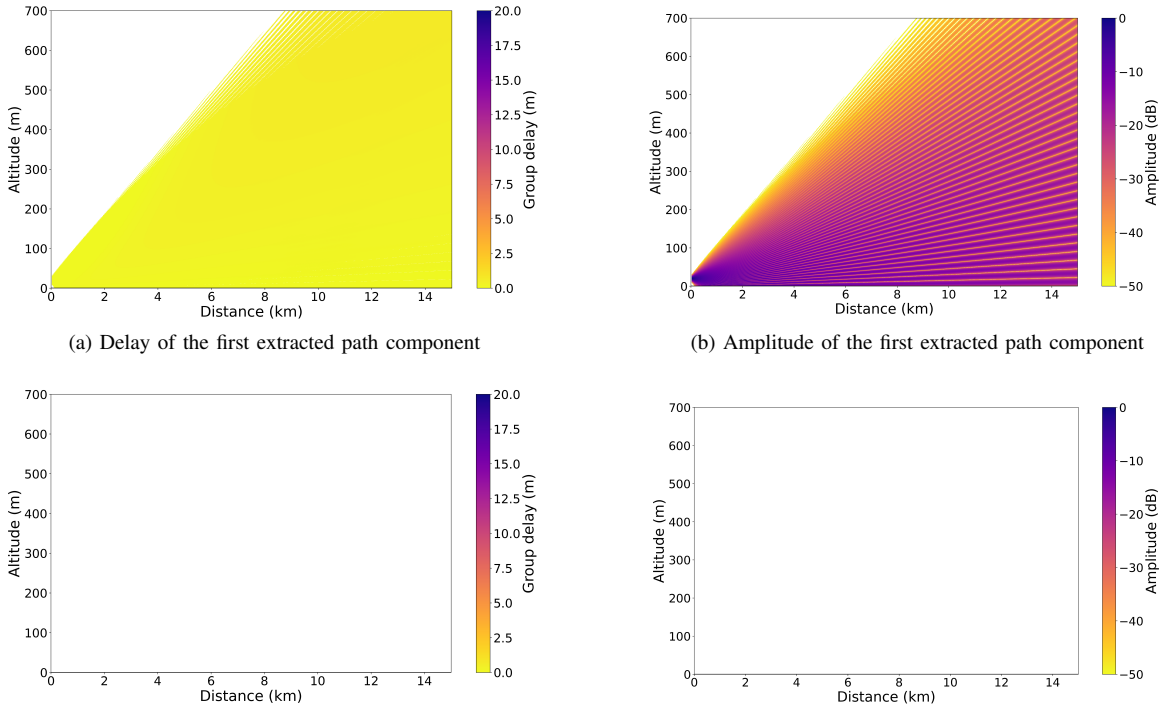
A simulation considering several large high-rise buildings is conducted, which can be modeled using vertical screens [36]. The urban environment is modeled using a randomly generated building-like terrain profile. A total of $N_{\mathrm{bu}} = 8$ buildings are randomly distributed along the horizontal axis, with each building's position independently selected. As the typical width of high-rise buildings (approximately 30-100 m) is much smaller than the simulation grid spacing ($\Delta x = 500$ m), buildings are approximated as vertical screens. Building heights are randomly sampled from the interval $[0, z_{\max_\mathrm{bu}}]$, with $z_{\max_\mathrm{bu}} = 400$ m. The distribution of vertical screens is represented by the black line in Fig. 14. The simulation incorporates the standard atmospheric model while accounting for the Earth's curvature. Details of the source configuration and simulation parameters are provided in Table IV.

The simulation results, displayed in Fig. 14 and Fig. 15, reveal distinct propagation behaviors. Consistent with the previous simulations, the first component is predominantly associated with direct wave propagation, as the electromagnetic waves travel along a direct path unobstructed by urban structures. The second component corresponds to waves influenced by reflections from the ground surface. For the second component, in addition to ground reflections, there are also components of waves diffracted by the screen, whose propagation directions are nearly parallel to the ground. The third and fourth components capture reflection and diffraction phenomena at the edges of high-rise buildings. This behavior results from wave obstruction caused by the irregular geometries and the spatial distribution of urban structures.

The extracted multipath components, shown in Fig. 15, illustrate the capability of the extraction method to disentangle multipath components, providing a understanding of wave behavior in environments with high obstacles.

V. CONCLUSION

This study has investigated the impact of complex environments on CNS signal propagation, highlighting the need for accurate simulation and effective separation of multipath



(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

Fig. 8. Extracted multipath components in propagation over a planar PEC ground in a homogeneous atmosphere (Source height: 20 m).

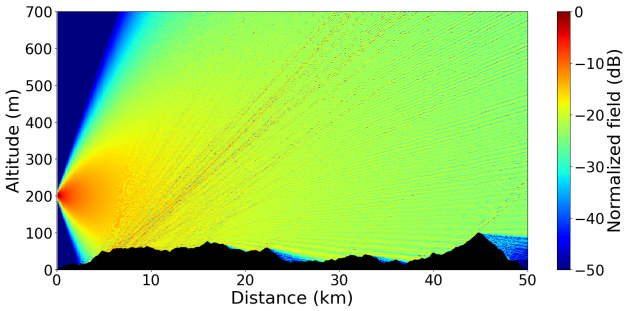


Fig. 9. Normalized electromagnetic wave field with SSW in propagation over an irregular dielectric ground in a homogeneous atmosphere.

TABLE IV
SOURCE AND SIMULATION PARAMETERS FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION D.

Parameters	Values	Parameters	Values
f	2 GHz	$z_{\max}$	1000 m
Polarization	TE	$z_{\max_cd}$	700 m
W_0	0.5 m	Δ_z	0.2 m
x_{w0}	-100 m	N_z	5000
y_s	0 m	$x_{\max}$	25 km
z_s	10 m	Δ_x	50 m
n_{f}	8	N_x	500
δ_{f}	1 MHz	n_{p}	4

components. Two split-step simulation methods have been introduced for modeling electromagnetic wave propagation in the lower troposphere, effectively accounting for inhomogeneous atmospheric conditions, ground impedance, and irregular terrain. A multipath extraction method was subsequently proposed, relying on Prony's method to estimate the amplitude, phase, and delay of multipath components from simulation results at only a few frequencies. The accuracy and numerical performance of the method have been evaluated, and a post-processing strategy has been suggested to enhance the interpretability of the results. To validate the method, simulations have been conducted under various environmental conditions. A comparison with the classical two-ray model with planar ground condition confirmed its accuracy. In that case, the extracted multipath components successfully reproduced the simulated channel, with a root mean square error below –50 dB across all spatial locations. However, a limitation is that the method performs less effectively when the delay spread between multipath components is small.

The proposed extraction approach is compatible with other frequency-domain electromagnetic simulation tools and can be extended to additional scenarios, such as urban canyons. It may also be adapted for use with two-way parabolic equation methods, particularly in configurations involving diffraction over vertical structures. Future research will involve broader comparisons with alternative extraction techniques in more complex environments, as well as validation through experimental measurements.

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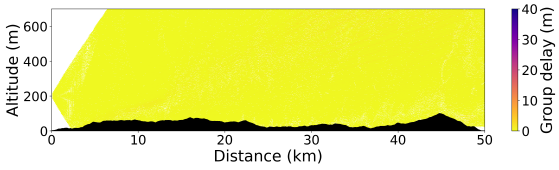
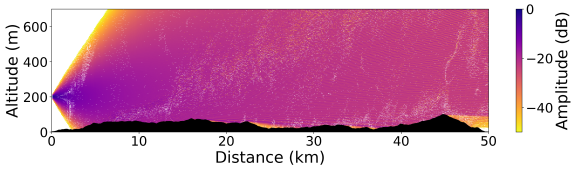
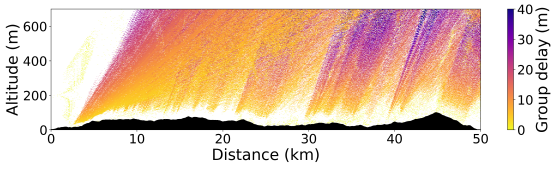


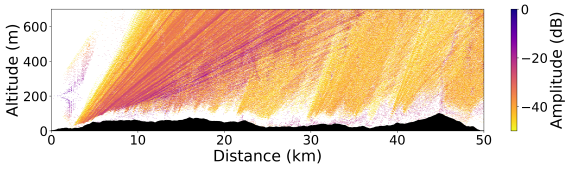
(a) Delay of the first extracted path component



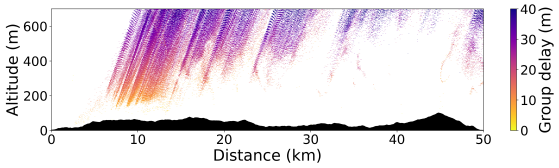
(b) Amplitude of the first extracted path component



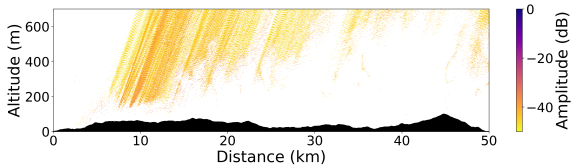
(c) Delay of the second extracted path component



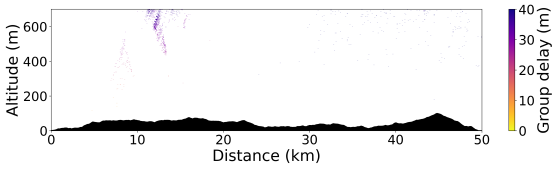
(d) Amplitude of the second extracted path component



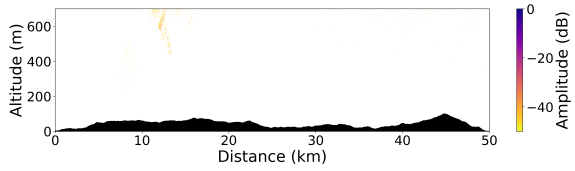
(e) Delay of the third extracted path component



(f) Amplitude of the third extracted path component



(g) Delay of the fourth extracted path component



(h) Amplitude of the fourth extracted path component

Fig. 10. Extracted multipath components in propagation over an irregular dielectric ground in a homogeneous atmosphere.

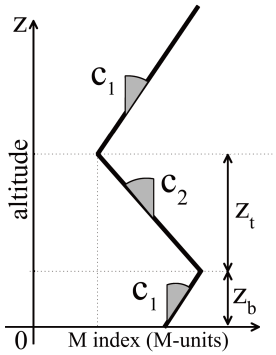


Fig. 11. Atmospheric modified refractivity profile.

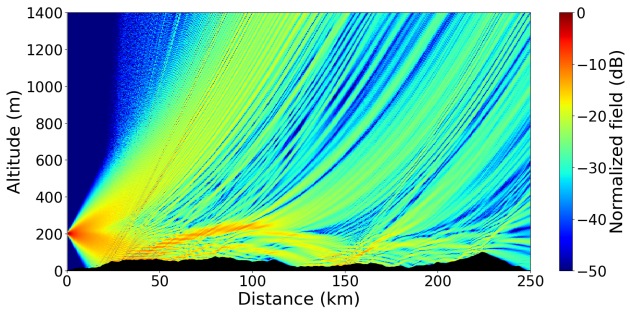


Fig. 12. Normalized electromagnetic wave field with SSW in propagation over an irregular dielectric ground within an atmospheric duct.

the Central Universities - Civil Aviation University of China Special Project (Grant No. 3122020075).

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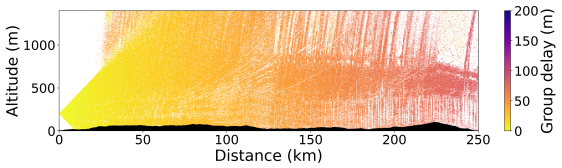
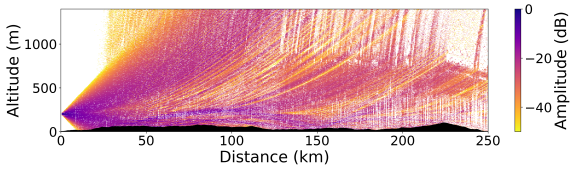
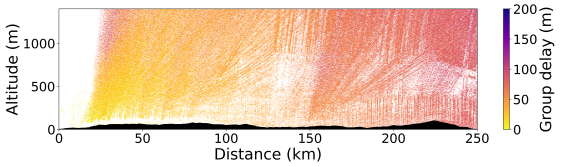


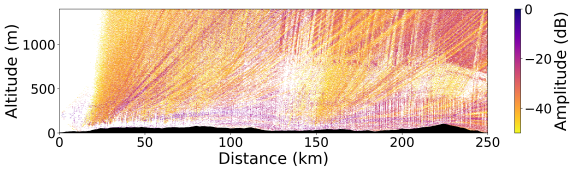
(a) Delay of the first extracted path component



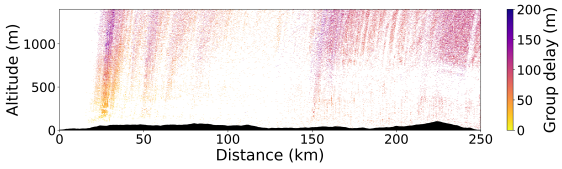
(b) Amplitude of the first extracted path component



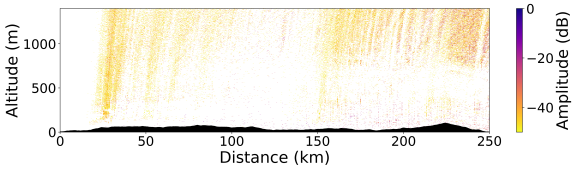
(c) Delay of the second extracted path component



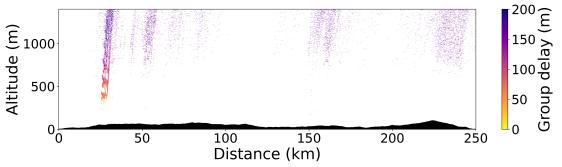
(d) Amplitude of the second extracted path component



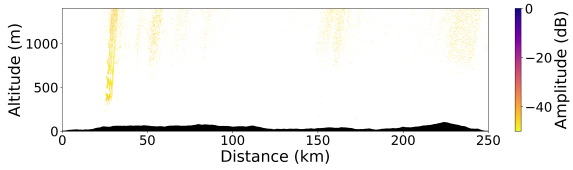
(e) Delay of the third extracted path component



(f) Amplitude of the third extracted path component



(g) Delay of the fourth extracted path component



(h) Amplitude of the fourth extracted path component

Fig. 13. Extracted multipath components in propagation over an irregular dielectric ground within an atmospheric duct.

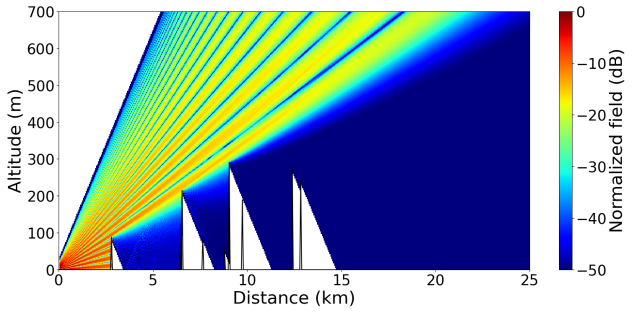


Fig. 14. Normalized electromagnetic wave field with SSW over vertical screens in a standard atmosphere (values below -50 dB are shown in white).

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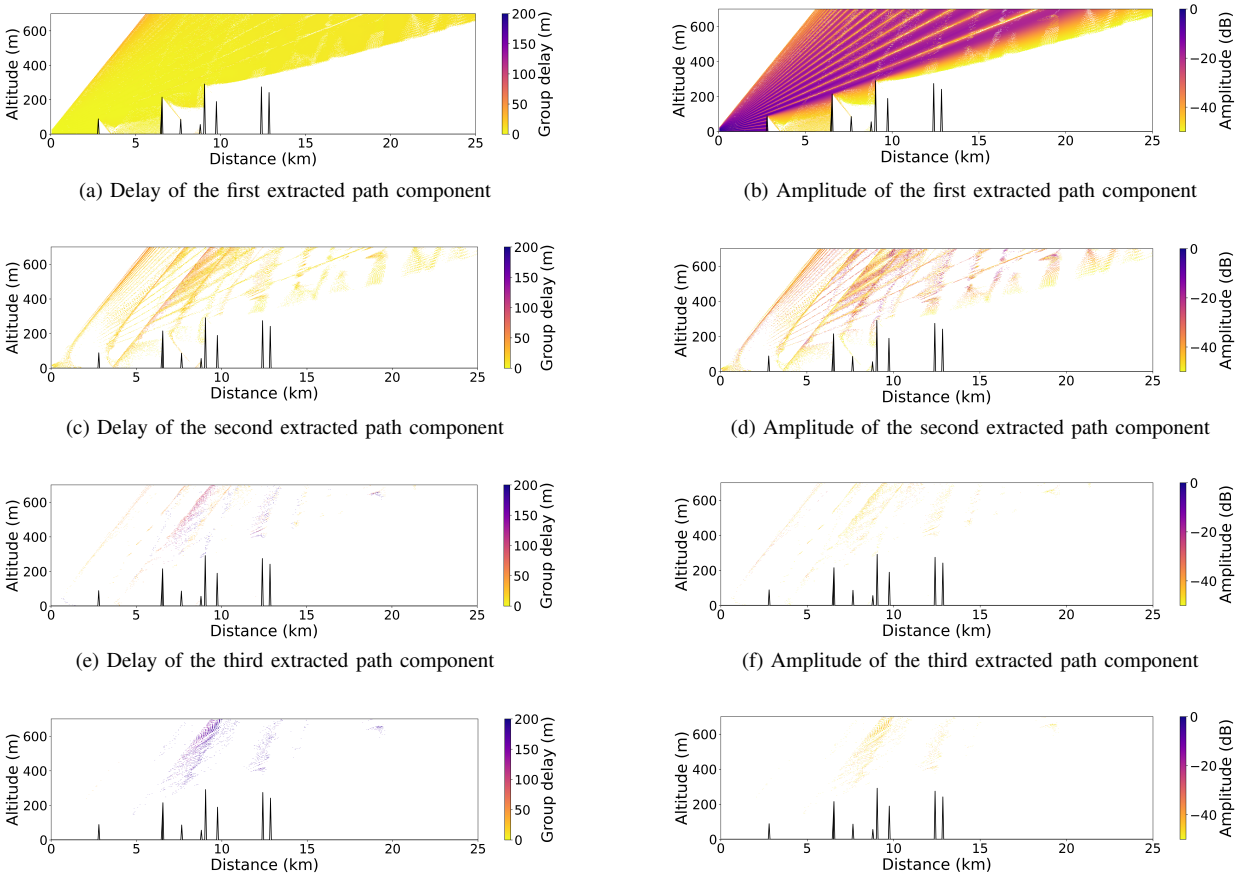

(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

(e) Delay of the third extracted path component

(f) Amplitude of the third extracted path component

(g) Delay of the fourth extracted path component

(h) Amplitude of the fourth extracted path component

Fig. 15. Extracted multipath components in electromagnetic wave propagation over vertical screens in a standard atmosphere.

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First of all, we sincerely thank the editor and six reviewers for their valuable and insightful comments. Below, we provide detailed responses to each comment. Corresponding modifications and additions have been made to the manuscript. They are highlighted in blue for clarity.

Reviewer 1

A.1 The abstract lacks specific information about key techniques used (e.g., Prony's method, post-processing).

Proposed modification: We appreciate the reviewer's suggestion. To address this, the following sentence has been added :
"We propose a multipath extraction approach to separate dominant propagation paths and associated multipath components by means of the post-processing of frequency-domain simulation results. The multipath extraction relies on the Prony's method, which allow to assess the multipath amplitude, phase, and delay from simulation results at a few frequencies."

A.2 No quantitative performance metrics or numerical results are provided in the abstract

Proposed modification: We agree with this comment. The following sentence has been added in the abstract:
"Simulation results are consistent with the two-ray model over flat ground. In this case, the extracted multipath reproduces the simulated channel with a root mean square error below –50 dB at all spatial points."

A.3 The application scope (e.g., UAM, radar systems) is not clearly articulated.

Response: The application scope of the article is indeed mainly on navigation and radar systems for UAV and UAM. Extracting delays in the multipath is important in that case. For radar systems, paths whose delays differ significantly from the direct path yield spurious echoes. For navigation systems where the time of arrival or the time-difference of arrival are used, delayed multipath impairs the positioning accuracy. Besides, UAV and UAM are in the very low troposphere, which is the domain where the environment (atmosphere and relief) is most likely to produce multipath.

Proposed modification: The following paragraph has been added to the third paragraph of the Introduction:
"Therefore, it is of primary importance to develop air-ground propagation channel models that are consistent with existing and future navigation and radar systems for unmanned aerial vehicle (UAV) and UAM. Extracting delays in the multipath is important in that case. For radar systems, paths whose delays differ significantly from the direct path yield spurious echoes. For navigation systems where the time of arrival or the time-difference of arrival are used, delayed multipath impairs the positioning accuracy. Besides, UAV and UAM operate in the very low troposphere, which is the domain where the environment (atmosphere and terrain) is most likely to produce multipath."

A.4 The importance of separating multipath components is not justified.

Response: Please refer to our response to the previous comment where we address the importance of separating multipath components notably for radar and navigation systems.

A.5 The introduction does not highlight the paper's novelty or contributions in relation to prior work.

Response: Relevant references have been added to the introduction, and the article's objective has been revised to better highlight its novelty.

Proposed modification: To address this comment, the following paragraph has been added to the second-to-last paragraph of the Introduction:
"The objective of this article is to assess the multipath characteristics of the low-troposphere propagation channel. The novelty of this work lies in combining split-step methods for computing the fields at a small number of frequencies with an efficient method for extracting multipath components in terms of amplitude, phase, and delay. Several numerical experiments are conducted, and the flat-ground scenario is validated against the two-ray model."

A.6 No discussion or citation of alternative methods such as parametric models, time-domain inversion, or ML-based extraction techniques.

Response: We thank the reviewer for this valuable suggestion. A new paragraph has been added to the eighth paragraph of the Introduction, discussing and citing alternative methods for multipath extraction, including parametric models, time-domain inversion techniques, and machine learning–based approaches. References [20]–[28] have been newly included.

Proposed modification: The following paragraph has been added in the 8th paragraph of the Introduction:
"To extract multipath components, signal processing techniques can be broadly categorized into three classes. Parametric methods, such as Prony's method [20], [21], the Matrix Pencil method [22], MUSIC, and ESPRIT [23], model the received signal as a sum of complex exponentials. These methods offer high-resolution estimation of multipath parameters, including path delays and amplitudes. However, techniques like MUSIC and ESPRIT typically require a large number of frequency samples and are optimized for scenarios with additive noise. Time-domain inversion techniques aim to reconstruct impulse responses or estimate multipath parameters directly from observed signals. Representative approaches include deconvolution, sparse recovery, matching pursuit, and CLEAN-based algorithms [24], [25]. In particular, IFFT-based reconstruction provides an efficient way to convert frequency-domain data into approximate time-domain responses. While these methods are computationally efficient and easily interpretable, their resolution is fundamentally constrained by the signal bandwidth and frequency sampling density. Machine learning – based methods, including convolutional neural networks (CNNs), graph-based models, and transformers, have also been applied to extract propagation parameters from electromagnetic simulation data [26]-[28]. These data-driven approaches exhibit strong generalization capabilities across complex environments. However, they typically require large volumes of labeled training data and may lack physical interpretability. Overall, these three categories of techniques offer complementary strengths for analyzing and extracting multipath effects in electromagnetic wave propagation, each suited to different application contexts and data characteristics."

A.7 The research gap is not clearly defined, and prior relevant journal work is missing.

Response: As explained in the previous responses, the objective of the article has been rewritten to more clearly highlight the research gap. In addition, more relevant journal references have been included.

B.II.1 Only SSW is used in experiments, but both SSF and SSW are introduced without clarifying their roles.

Response: SSF is a well-established canonical approach for simulating electromagnetic wave propagation in the lower troposphere. SSW is a more recent alternative to SSF, offering improved computational efficiency while maintaining comparable simulation accuracy. In the experiments, multipath components are extracted exclusively from the SSW simulation results. We have verified that the extracted components are consistent with those obtained using SSF.
For more details on the SSF and SSW methods, please refer to the following previously published works:
[1] H. Zhou, R. Douvenot, and A. Chabory, "Modeling the long-range wave propagation by a split-step wavelet method," *J. Comput. Phys.*, vol. 402, p. 109042, Feb. 2020.
[2] T. Bonnafont, R. Douvenot, and A. Chabory, "A local split-step wavelet method for the long range propagation simulation in 2D," *Radio Sci.*, vol. 56, no. 2, p. 1-11. 2021.

Proposed modification: At the end of Section II. C, the following sentence has been added: "Previous studies [14], [19] have shown that SSW maintains the accuracy of SSF with reduced computation time. For this reason, in this article, only simulations with SSW are considered."

B.II.2 No justification is provided for simulation design choices (e.g., frequency sampling, wavelet compression) and how they support multipath extraction.

Response: Further explanation has been added to justify the frequency sampling scheme and the wavelet compression threshold used in the simulations.

Proposed modification: In the third and fourth paragraphs of Section IV.A, the following text has been added:
- "The compression threshold in SSW is set to 10^{-8}, which corresponds to a maximum compression error of -100 dB, as described in [14], [34]."
- "According to (14), δ_f is set to 0.5 MHz, so that multipath with delays up to 600 m can be extracted without ambiguity. This results in a sequence of frequencies: 5.0000, 5.0005, 5.0010, and 5.0015 GHz."

B.II.3 The section lacks any stated optimization, variation, or novel integration of SSW into the extraction workflow.

Response: As introduced in the article, the proposed multipath extraction method is compatible with simulation results from any high-accuracy electromagnetic propagation method. The SSW method is particularly suitable for simulating long-range electromagnetic wave propagation at low altitudes. Therefore, it has been chosen in this study to support the extraction process.

B.III.1 The novelty of applying Prony's method to simulation-generated frequency-domain data should be explicitly stated and contextualized.

Response: The objective of the article has been rewritten to better underline the novelty.

Proposed modification: Please refer to the revised text introduced in response to comment A.5.

B.III.2 No justification is given for post-processing parameters such as a_{thres} and τ_{min}, a sensitivity analysis is recommended.

Response: We appreciate the reviewer's observation and provide below the rationale for the choice of the post-processing parameters.

Proposed modification: In the third and fifth paragraphs of Section III.D, the following text has been added:

a_{thres} : "Due to various uncertainties (e.g., atmospheric refraction index, terrain relief, ground composition) and simulation approximations, the channel response is only approximately modeled. Therefore, in numerical experiments, the threshold below which paths are removed is set to -60 dB."

τ_{min} : "The choice of τ_{min} depends on the parameters of the system under study, notably its bandwidth. Moreover, for radar and navigation systems, the delay resolution typically corresponds to a path length of at least 1 m. Therefore, τ_{min} is set to $1/c_0$, where c_0 denotes the speed of light in vacuum. [30], [31]"

B.III.3 Equation (15) includes overlapping conditions ($\geq$ and $\leq$) that create logical inconsistencies and need correction.

Proposed modification: We agree with the reviewer's comment. The overlapping conditions in (15) have been corrected to use strict inequalities: the first condition is now "$>$" and the second condition remains "$\leqslant$".

B.IV.1 Section IV(A): Some variables (e.g., xw0, k) are used without proper definition. The rationale for using CSP is not provided.

Response: The definitions related to the complex source point have been provided in the article (Section IV.A). Any electromagnetic wave can be decomposed into a linear combination of Gaussian waveforms at different scales. Since the electromagnetic wave emitted by the CSP is Gaussian-shaped, the proposed method is effective for CSP-emitted signals and, by extension, applicable to all electromagnetic waves.

Proposed modification: In the first paragraph of Section IV.A, the following text has been added:
"CSP represents an antenna whose radiation pattern is approximately Gaussian. This could correspond to a Potter or corrugated horn antenna. Moreover, several methods exist to expand any radiation into a linear combination of Gaussian waveforms. Therefore, by extension, the method is applicable to any electromagnetic waves, including those characterized by an arbitrary CNS antenna radiation pattern."

B.IV.2 In Figure 7: Domain height is inconsistent (e.g., 800 m vs. 1000 m); the usable simulation region shrinks due to a fixed 30% absorbing layer. This change should be stated and justified.

Proposed modification: In the numerical experiment of Section IV.B, the total simulation domain height has been updated to z_{max} = 1000 m. After the absorbing layer is omitted from the figures, the effective computational domain height becomes z_{max_cd} = 700 m. All relevant figures have been revised accordingly (now Figures 9 and 10 in the updated manuscript). Additionally, the absorbing layer is now illustrated and discussed only in

Figure 2, as suggested by Reviewer 3.
In Tables II, III, and IV, z_{max_cd} has been added to indicate the maximum altitude of the effective computation domain, which matches the vertical extent shown in the figures.
In the third paragraph of Section IV. A, the following text has been added:
"In Table II, z_{max_cd} denotes the maximum altitude of the effective computational domain, excluding the absorbing layer."
In the fifth paragraph of Section IV. A, the following text has been added:
"The upper zone above the dashed line corresponds to the apodization/absorbing layer. In this simulation, this zone is shown for illustration purposes only and is omitted in the results of the other simulations for simplicity."

B.IV.3 Post-processing thresholds are applied without justification or explanation of their impact.

Response: We acknowledge the reviewer's concern regarding the justification of post-processing thresholds. Please refer to our response to the comment B.III.2.

B.IV.4 Section IV(C): No equation or full parameter definition is given for the atmospheric duct model.

Response: The atmospheric duct model used in this work is a classical model detailed in:
- M. Levy, *Parabolic Equation Methods for Electromagnetic Wave Propagation*, London, U.K.: IET, 2000.

As presented on page 83 of Levy's book, the ducting layer in the atmosphere can often be represented by a simple trilinear profile, which is also provided there.

Proposed modification: In the first paragraph of Section IV.C, the following text has been added:
"The atmospheric model, illustrated in Fig. 11, is defined by a trilinear profile commonly used to represent ducting layers in the atmosphere [13]. The model is characterized by the following parameters: z_b = 200 m and z_t = 1000 m, which denote the bottom and top heights of the ducting layer, respectively; and c_1 = -0.1 M-unit/m, c_2 = 0.118 M-unit/m, which represent the vertical gradients of modified refractivity within different sublayers."

B.IV.5 Section IV(D): Vertical screen parameters (height, spacing, number) are not provided, Earth curvature is mentioned but not modelled or explained, the urban layout (building shapes, heights, ground variation) is not described, no explanation or interpretation is given for extracted components in Figure 13, no justification of (e.g., frequency, spacing, domain size) change across experiments.

Response:
(1) Details regarding the vertical screen parameters have been added to clarify the simulation setup.
(2) The effect of Earth curvature is a classical modeling consideration, as discussed in Levy's book (Section 4.4: Earth Flattening Transformation, pp. 53–56). In our model, this effect is incorporated into the modified refractive index, consistent with established propagation modeling practices. We agree that this should be mentioned in the article.
(3) An explanation of the extracted components in Figure 13 (now Figure 15 in the revised manuscript) has been added to clarify their interpretation and relevance.
(4) To demonstrate the general applicability of the proposed method, simulation results are provided under various parameter settings. As such, parameter values differ between scenarios in order to showcase robustness under different conditions.

Proposed modification:
(1) In the first paragraph of Section IV.D, the following text has been added:
"The urban environment is modeled using a randomly generated building-like terrain profile. A total of N_{bu}= 8 buildings are randomly distributed along the horizontal axis, with each building's position independently selected. As the typical width of high-rise buildings (approximately 30-100 m) is much smaller than the simulation grid spacing (Δ_x = 500 m), buildings are approximated as vertical screens. Building heights are randomly sampled from the interval $[0, z_{max_bu}]$, with z_{max_bu} = 400 m. The distribution of vertical screens is represented by the black line in Fig. 14."
(2) In the first paragraph of Section IV.C, the following text has been added:
"In this article, the Earth curvature is realized by using modified refraction index, as introduced in [13]."
This approach is applied consistently in both simulations C and D.
(3) In the second paragraph of Section IV.D, the following text has been added:
"For the second component, in addition to ground reflections, there are also components of waves diffracted by the screen, whose propagation directions are nearly parallel to the ground."

B.V.1 The conclusion does not highlight the originality or methodological contribution of the work.

Response: To address all the comments from B.V.1 to B.V.4, the conclusion has been revised to better emphasize the originality, methodological contributions, key results, limitations, and potential future work of the study.

Proposed modification: The conclusion (Section V) has been rewritten as follows:
"This study has investigated the impact of complex environments on CNS signal propagation, highlighting the need for accurate simulation and effective separation of multipath components. Two split-step simulation methods have been introduced for modeling electromagnetic wave propagation in the lower troposphere, effectively accounting for inhomogeneous atmospheric conditions, ground impedance, and irregular terrain. A multipath extraction method was subsequently proposed, relying on Prony's method to estimate the amplitude, phase, and delay of multipath components from simulation results at only a few frequencies. The accuracy and numerical performance of the method have been evaluated, and a post-processing strategy has been suggested to enhance the interpretability of the results. To validate the method, simulations have been conducted under various environmental conditions. A comparison with the classical two-ray model with planar ground condition confirmed its accuracy. In that case, the extracted multipath components successfully reproduced the simulated channel, with a root mean square error below –50 dB across all spatial locations. However, a limitation is that the method performs less effectively when the delay spread between multipath components is small.

The proposed extraction approach is compatible with other frequency-domain electromagnetic simulation tools and can be extended to additional scenarios, such as urban canyons. It may also be adapted for use with two-way parabolic equation methods, particularly in configurations involving diffraction over vertical structures. Future research will involve broader comparisons with alternative extraction techniques in more complex environments, as well as validation through experimental measurements."

B.V.2 Key results (e.g., extraction accuracy) and methodological benefits are not referenced.

Response: Thank you for this comment. The conclusion section has been revised to explicitly highlight key results. Please refer to the comment B.V.1.

B.V.3 Limitations of the current approach are not discussed.

Response: To address the limitations of the current approach, an additional simulation has been conducted based on the setup in Section IV.A, with the source height reduced to 20 m to examine its impact on multipath separation. The results show that when the delay spread is short, the method is unable to reliably distinguish multiple paths.

Proposed modification: In the last paragraph of Section IV.A, the following text has been added:
"When the source height is set to 20 m, the delay between the reflected and direct paths is less than 2 m. The normalized propagated field is illustrated in Fig. 7. After applying the multipath extraction method, the first and second components are presented in Fig. 8. The amplitude of the second component is negligible (under -50 dB). In this case, only one dominant path is sufficient to characterize the channel. This shows a limitation of the method: it can effectively separate different components only when there is a sufficient delay between them."
Additionally, this limitation is mentioned in the revised conclusion. Please refer to the response to comment B.V.1.

B.V.4 Future work directions should be added to increase impact.

Response: Future research directions have been explicitly included in the revised conclusion. Please refer to the comment B.V.1.

C.1. Figures should be referenced and explained in the text before they are presented (e.g., Figures 2, 3, 6, 7).

Proposed modification: The positioning of the figures has been improved.

C.2. Add panel labels (a), (b), etc., directly into figures.

Proposed modification: As suggested, all subfigures now include panel labels (a), (b), etc.

C.3. Improve caption clarity and consistency across all Figures.

Proposed modification: The clarity and consistency of all figure captions have been reviewed and improved to ensure uniform formatting and better comprehension.

C.4. Figure 3:
Ensure consistent subplot layout:
- Row 1: (a) Direct path – Delay (left), Amplitude (right)
- Row 2: (b) Reflected path – Delay (left), Amplitude (right)

Align Y-axes and axis ticks, and add clear panel labels within the figure, not just in the caption.
Caption clarity, Example: Figure 3. Extracted multipath components in propagation over a planar PEC ground. (a) Group delays for direct and reflected paths. (b) Corresponding field amplitudes.

Proposed modification: Figure 3 has been improved according to the reviewer's comment.

C. 5. Figure 4:
Missing X-axis units in (b).
Caption lacks panel references (a), (b).

Typo “2rd” should be corrected to “2nd”, and the term "2nd part" is unclear—should be defined or removed.

Proposed modification: The x-axis label in panel (b) has been updated to include the unit (dB). The typo "2rd part" has been corrected to "2nd path".

C. 6. Figure 7: The ground profile is shown but not described. Is it synthetic, real DEM, or randomly generated?

Proposed modification: In the revised manuscript, this figure is now labeled as Figure 9. To clarify the nature of the terrain, we have added the following text in the first paragraph of Section IV.B:
“We adopt a terrain generation method in which the terrain profile is formed by the superposition of random surfaces defined across multiple scales [35]. At each scale, random height values are sampled and interpolated, typically using linear or cosine interpolation, to create smooth transitions. The resulting multi-scale layers are then weighted and summed to produce a one-dimensional terrain that exhibits both randomness and natural continuity.”

C.7. Figure 8: No labels or description of which row corresponds to which multipath component.
Caption should explicitly identify components: e.g., (a) direct, (b) reflected, etc.

Proposed modification: In the revised manuscript, this figure is now labeled as Figure 10. The figure has been improved according to the reviewer’s comment.

C.8. Figures 11 & 13:
Suffer from the same issues as Figure 8.

Proposed modification: In the revised manuscript, these figures are now labeled as Figure 13 and Figure 15. Both figures have been improved according to the reviewer’s comment.

C.9. Figure 9: The caption should be updated to: “Atmospheric modified refractivity profile” or “Surface-based duct model,” instead of “Atmospheric duct model of refractive index.”

Proposed modification: In the revised manuscript, this figure is now labeled as Figure 11. The caption has been updated to “Atmospheric modified refractivity profile”.

D.1 The reference list includes too many conference papers; add more peer-reviewed journal articles, especially from IEEE Transactions.

Proposed modification: The references suggested by the reviewers have been included in the bibliography, and all of them are peer-reviewed journal articles. In addition, new references from reputable journals, especially IEEE Transactions, have been added to strengthen the scholarly foundation of the article.

D.2 No direct citation is provided for Prony’s method, a core part of the methodology.

Proposed modification: To address the reviewer’s comment, we have added direct citations supporting the use of Prony’s method in both the Introduction and Section III.B. The following references have been included:
[1] K. Sayidmarie and S. Shanshal, "Prediction of ground multipath parameters from range-gain measurements using Prony's technique," *IET Microwaves, Antennas & Propagation*, vol. 3, no. 7, pp. 1127–1132, 2009.

[2] L. Weiss and R. N. McDonough, "Prony ' s method, Z-transforms, and Padé approximation," *SIAM Rev.*, vol. 5, no. 2, pp. 145–149, Apr. 1963.
(This is the original publication of Prony's method)

D.3 Important recent works (last 3–4 years) on simulation-based multipath extraction and frequency-domain estimation are missing.

Proposed modification: To address this comment, we have updated the bibliography by incorporating the references suggested by the reviewers as well as additional important recent works.

Reviewer 2

G.1 This manuscript mainly focuses on the application of air-ground links (such as UAM), which generally have large elevation or depression angles. However, the examples abolished in the paper calculations are mostly zero degree elevation angles (meaning horizontal links).

Response: We thank the reviewer for the comment. According to NASA's Urban Air Mobility Concept of Operations, UAM aircraft operating at cruising altitudes generally between 1,200 and 5,000 feet above ground level (AGL). These altitude ranges imply that UAM primarily operates at relatively low elevation angles when communicating with ground-based CNS systems. As a result, they are more vulnerable to multipath propagation effects caused by atmospheric ducting, surface reflections, and terrain irregularities. In contrast, for near-zenith propagation paths, the risk of multipath delays is small.

Proposed modification: In the 7th paragraph of Section I, the following text has been added:
"NASA's Urban Air Mobility Concept of Operations [16] envisions UAM aircraft operating at cruising altitudes generally between 1,200 and 5,000 feet above ground level (AGL). These altitude ranges imply that UAM primarily operates at relatively low elevation angles when communicating with ground-based CNS systems. As a result, they are more vulnerable to multipath propagation effects caused by atmospheric ducting, surface reflections, and terrain irregularities. In contrast, for near-zenith propagation paths, the risk of multipath delays is small."

G.2 The parabolic equation has a certain range of applicable angles. The method in this article needs to provide its applicable range of angles and whether it is a wide-angle parabolic equation.

Response: As the reviewer pointed out, the standard parabolic equation method involves a paraxial (small-angle) approximation. In our work, we employ a wide-angle parabolic equation. The range of applicable angles for our implementation is approximately –45° to +45°.

Proposed modification: In Section II.B. Step 2), the following sentence has been added:
"The wide-angle parabolic equation is applied with the range of applicable angles is approximately –45° to +45°."

G.3 As for extracting multipath information, ray tracing may be a more direct method. Could it be compared with the method proposed in this manuscript?

Response: A comparison between the proposed method with ray tracing has been performed for the planar ground scenario in Section IV.A. In future work, we will extend the comparison of the proposed method with other approaches, such as ray tracing, in more complex scenarios.

Proposed modification: In the conclusion, the following sentence has been added:
"In future work, we will extend the comparison of the proposed method with other approaches in more complex scenarios."

S1. Page 2, Col.2, Line 17, it is said that 'An inhomogeneous atmosphere is assumed', however, there are no information in detail.

Response: In both the SSF and SSW methods, an inhomogeneous atmosphere is considered. Specifically, a non-uniform refractive index profile is introduced through the

use of range-dependent phase screens, a classical technique in parabolic equation-based propagation modeling. This approach allows for the simulation of vertical refractive gradients typically encountered in the lower troposphere.

S2. Sec.2, The initial antenna pattern was not provided, which were needed for calculation of PE.

Proposed modification:
(1) In the first paragraph of Section III.B, the following text has been added:
"The simulation starts at x = 0 where the field is assumed to be known. This field is generally obtained either from the transmitting antenna pattern, or from an equivalent radiating aperture [13]. The propagation is then computed stepwise from x to $x + \Delta x$ in the subsequent steps:"

(2) In the first paragraph of Section IV.A, the following text has been added:
"CSP represents an antenna whose radiation pattern is approximately Gaussian. This could correspond to a Potter or corrugated horn antenna. Moreover, several methods exist to expand any radiation into a linear combination of Gaussian waveforms. Therefore, by extension, the method is applicable to any electromagnetic waves, including those characterized by an arbitrary CNS antenna radiation pattern."

S3. Page 5, Col.2, Line 40, it is not clear how to obtain 'the reconstructed fields'.

Proposed modification: The reconstructed fields are obtained using (4). A reference to equation (4) has been added in the corresponding sentence to clarify how the reconstructed fields are obtained.

Reviewer 3

1. The absorbing layer is an implementation detail of the numerical method and should not be illustrated in the examples in this article. It may mislead unprepared readers and generate unnecessary questions.

Proposed modification: The apodization layer is illustrated and discussed solely in Figure 2 and omitted from all other simulations.
In Tables II, III, and IV, z_{max_cd} has been added to indicate the maximum altitude of the effective computation domain, excluding the absorbing layer. This height matches the vertical extent shown in the corresponding figures.
In the third paragraph of Section IV. A, the following text has been added:
"In Table II, z_{max_cd} denotes the maximum altitude of the effective computational domain, excluding the absorbing layer."
In the fifth paragraph of Section IV. A, the following text has been added:
"The upper zone above the dashed line corresponds to the apodization/absorbing layer. In this simulation, this zone is shown for illustration purposes only and is omitted in the results of the other simulations for simplicity."

2. How can the required number of paths (n_p) be estimated in advance?

Proposed modification: We thank the reviewer for this insightful comment.
In the second paragraph of Section III.A, the following text has been added:
"A limitation of all parametric multipath estimation methods (Prony's method, ESPRIT, MUSIC, etc.) is that the number of propagation paths n_p must be specified in advance. However, this value is generally not known a priori and cannot be reliably determined without prior knowledge of the channel structure. In typical ground-based line-of-sight propagation scenarios, the number of dominant multipath components is generally low due to the limited number of significant reflectors near the ground. Based on prior studies, we expect that a small number of paths (usually fewer than 5) are sufficient to capture the essential multipath structure. "

3. What is the computational complexity of the method? How does the computation time compare to a standard PE approach (without multipath estimation)?

Proposed modification: We thank the reviewer for this valuable comment. A new section on complexity analysis has been added (Section III. E):
"*E. Complexity analysis*
The total computational complexity of the SSW method combined with the multipath extraction method, denoted as C_{TC}, can be expressed as

$$C_{TC} = n_f C_{SSW} + n_x n_z C_{ext}$$

where

- C_{SSW} denotes the computational cost of the SSW method. As detailed in [32], the complexity of the fast wavelet transform and the associated compression operations is approximately $O(N_x N_z)$.
- C_{ext} represents the complexity of the multipath extraction at a single point. This includes solving (11), computing the roots of a polynomial of degree n_p, and solving (8), a linear system of size n_f.

In the numerical experiments, only a limited number of frequencies n_f, and hence a small number of propagation paths n_p, are extracted, the overall computational complexity is generally dominated by $n_f C_{SSW}$."

4. Since you are considering diffraction on vertical obstacles, experiments for two-way PE should be conducted.

Response: We appreciate the reviewer's suggestion. The proposed extraction method is compatible with two-way parabolic equation (PE) models and could be applied similarly to account for diffraction over vertical obstacles. However, this extension involves additional modeling considerations and validations, and is therefore left for future work. This has been mentioned in the revised conclusion. Please also refer to our response to Reviewer 1 (comment B.V.1).

Reviewer 4

Major 1. The core technical method—application of Prony's method for delay estimation—is well known in signal processing literature and not novel. The paper would benefit from either: i) a comparative study with modern high-resolution techniques (e.g., MUSIC, ESPRIT), or ii) a formal justification for preferring Prony's method in this context (e.g., noise resilience, computational complexity, or limited frequency samples).

Response: We agree that Prony's method is a classical technique in signal processing.
The novelty of our work does not lie in the use of Prony's method alone, but rather in its integration with the SSW method to enable efficient multipath extraction (amplitude, phase, and delay) from a small number of frequency-domain simulation results in the context of low troposphere channels.
High-resolution methods such as MUSIC and ESPRIT typically require a large number of samples and are designed to operate in noisy environments. In contrast, the context of this study involves simulation-generated data at a small number of discrete frequencies, with no noise (aside from numerical errors whose characteristics do not match the usual additional white noise approximation).

Proposed modification:
(1) The following paragraph has been added to the second-to-last paragraph of the Introduction:
" The objective of this article is to assess the multipath characteristics of the low-troposphere propagation channel. The novelty of this work lies in combining split-step methods for computing the fields at a small number of frequencies with an efficient method for extracting multipath components in terms of amplitude, phase, and delay. Several numerical experiments are conducted, and the flat-ground scenario is validated against the two-ray model."
(2) For further discussion on MUSIC and ESPRIT methods, please refer to our response to Reviewer 1, Comment A.6.

2. The validation strategy is based exclusively on synthetic simulation scenarios. No experimental or measurement-based validation is provided, despite the existence of relevant references [4]–[6]. Furthermore, there is no discussion of performance under noise, non-idealities, or incomplete data—conditions that are expected in practical scenarios.

Response: The objective of this article is to develop a method for extracting multipath characteristics—amplitude, phase, and delay—directly from frequency-domain simulation results at a limited number of discrete frequencies. A comparison between the extracted multipath components and measurements obtained via channel sounding is considered a promising direction for future work.

Proposed modification: In the first paragraph of Section III.A, the following sentence has been added:
"In this study, no noise is considered, as the aim is to efficiently extract the multipath components in terms of amplitude, phase, and delay from simulation results using only a few frequency samples. The numerical errors inherent in the simulation cannot be appropriately modeled as additional Gaussian noise."
In addition, the objective stated in the Introduction has been revised. Please also refer to the response to Reviewer 1, Comment A.5 for further details.

3. The paper would benefit from an analysis of: sensitivity to noise and simulation errors, resolution limits (in terms of δf and np), phase ambiguity conditions and mitigation strategies.

Response:
We thank the reviewer for the valuable suggestion. Since our simulations are noise-free numerical results, noise sensitivity is not directly addressed in this work.
(1) Resolution limits: Please refer to our detailed responses to Reviewer 1, comment B.V.3, and Reviewer 3, comment 2, which discuss resolution limits with respect to the frequency sampling interval δf and the number of multipath components n_p.
(2) Phase ambiguity: This issue has been addressed in our recent work:
- H. Zhou and A. Chabory, "Extraction methods of the dominant delay in simulation of electromagnetic waves in the low troposphere," in *Proc. IEEE Int. Symp. Antennas Propag. USNC-URSI Radio Sci. Meeting*, accepted for publication, 2025.

Proposed modification: At the end of Section III.B, the following sentence has been added: "The phase ambiguity conditions are further discussed in [29]."

4. While the SSW method is said to be more efficient than SSF, no quantitative data (e.g., runtime, memory usage, scaling with Nx/Nz) is provided. This weakens claims of improved performance.

Response: All relevant details of the SSF and SSW methods are detailed in our previously published articles:
[1] H. Zhou, R. Douvenot, and A. Chabory, "Modeling the long-range wave propagation by a split-step wavelet method," *J. Comput. Phys.*, vol. 402, p. 109042, Feb. 2020.
[2] T. Bonnafont, R. Douvenot, and A. Chabory, "A local split-step wavelet method for the long-range propagation simulation in 2D," *Radio Sci.*, vol. 56, no. 2, pp.1-11, 2021.
These references have already been cited in the current manuscript.

5. The post-processing step (thresholding and delay clustering) is described in ad hoc terms, without formal criteria or sensitivity analysis. It would be helpful to define how thresholds athres and τmin are selected and how they affect results.

Response: Thank you for the comment. Please refer to our detailed response to comment B.III.2 from Reviewer 1, where the selection of a_{thres}, τ_{min} and their impact on the results are discussed.

Minor Comments

• Improve the explanation of the matrix formulation (Eqs. 5–8) with clearer definitions and notation.

Proposed modification: The explanation of the matrix formulation (Eqs. 5 – 8) has been improved.

• Clarify the assumptions behind Eq. (4), particularly the narrowband assumption and noise-free simulation.

Response: Thank you for your comment. The following clarifications have been added regarding the assumptions of Eq. (4).

(1) Noise-free assumption: Additive noise is not considered in this study. Numerical errors inherent in the simulation are not treated as noise.
(2) Bandwidth assumption: Equation (4) represents a wideband channel model. A reference to [F. Pérez-Fontán and P. Mariño, *Modeling the Wireless Propagation Channel*. Wiley, 2008.] has been added to support this formulation.

Proposed modification: In the first paragraph of Section III.A, the following sentence has been added:
" This equation represents a wideband channel model, as indicated in [8]."

• The conclusion could be strengthened by discussing limitations and potential extensions (e.g., real-time implementation, integration with CNS systems).

Response: Thank you for your valuable suggestion. Please refer to our detailed response to comment B.V.1 from Reviewer 1, where we discuss the limitations and potential extensions.

Typographical and Editorial Issues

• Pg. 3: "homogeneous difference equation" → could benefit from a brief reminder of the assumption (no noise).

Proposed modification: In the first paragraph of Section III.A, the following sentence has been added:
"In this study, no noise is considered, as the aim is to efficiently extract the multipath components in terms of amplitude, phase, and delay from simulation results using only a few frequency samples. The numerical errors inherent in the simulation cannot be appropriately modeled as additional Gaussian noise."

• Fig. 4 caption: "analytic" → should be "analytical".

Proposed modification: The caption has been corrected by changing "analytic" to "analytical" in accordance with the reviewer's comment.

• References should follow a consistent format (capitalization, punctuation, author order) and doi should be included.

Proposed modification: The references have been revised in accordance with the reviewer's comment.

Additional suggested references

To strengthen the manuscript's connection with current research and provide a more robust contextual framework, the authors are encouraged to include the following references. These works offer relevant insights into signal decomposition techniques, electromagnetic wave propagation modeling, tropospheric effects, and validation strategies—thereby helping to more clearly position the proposed approach within the broader scientific landscape:

• R. Schmidt, "Multiple emitter location and signal parameter estimation," in IEEE Transactions on Antennas and Propagation, vol. 34, no. 3, pp. 276-280, March 1986, doi: 10.1109/TAP.1986.1143830.
→ This foundational paper introduces the MUSIC algorithm, a high-resolution method for multipath and direction-of-arrival estimation, which could be cited for comparison with Prony's method.

• Y. Hua and T. K. Sarkar, "Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise," in IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 38, no. 5, pp. 814-824, May 1990, doi: 10.1109/29.56027.
→ Offers an alternative to Prony's method that is more robust in noisy environments. Citing it would provide methodological depth.

• T. Tunçkol, M. Biscarini and L. Luini, "Radiometric Estimation of Tropospheric Attenuation: A Mixed Physically Based/Machine Learning Approach," in IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1-13, 2024, Art no. 5301113, doi: 10.1109/TGRS.2024.3393506.
→Introduces a modern approach to modeling tropospheric attenuation through a hybrid physical/statistical technique that could complement the propagation scenarios considered in the manuscript.

• D. W. Matolak and R. Sun, "Air–Ground Channel Characterization for Unmanned Aircraft Systems—Part I: Methods, Measurements, and Models for Over-Water Settings," in IEEE Transactions on Vehicular Technology, vol. 66, no. 1, pp. 26-44, Jan. 2017, doi: 10.1109/TVT.2016.2530306
→ A key reference for realistic channel modeling in UAS applications, helping bridge simulation-based and empirical perspectives.

> **Proposed modification**: The references suggested by the reviewer have been added to the bibliography and cited at appropriate locations in the manuscript to strengthen the contextual grounding and highlight methodological comparisons.

Recommendation

Major Revision

The topic is relevant and the structure of the manuscript is clear, but the authors need to address the methodological originality, provide stronger validation (ideally with measured or noisy data), and better justify performance claims. I look forward to reviewing a revised and improved version of this work.

Reviewer: 5

Comments to the Author
This paper explores the application of Prony's algorithm to the extraction of multipath parameters for simulated tropospheric propagation channels.

Although I recommended for major revision, the paper itself (explanations, explanation of theories, etc.) is very good but I felt that the work needed a bit more.

The presentation of simulation methods actually precedes what the authors are stating is the main contribution of the paper, thus generating (at least for this reviewer) some confusion about the importance of the simulation method in this contribution versus the stated contribution (which is the extraction of the multipath parameters). The simulation methodology should, in my opinion, have followed the extraction approach to allow for the clear communication of the stated contribution first and the mechanisms for providing proof of concept discussed later in the paper.

> **Responses:** We thank the reviewer for the comment. This article presents a method for extracting multipath characteristics based on simulation results rather than measured data. Therefore, we first introduce the long-range electromagnetic wave propagation simulation methods in the lower atmosphere, followed by the multipath extraction approach that operates on these simulation outputs.
> Our intention was to present the state-of-the-art simulation techniques first, as they form the foundation of the proposed extraction method. Nevertheless, we are open to inverting Sections II and III if the editor considers this reordering more appropriate for emphasizing the main contribution.

I would recommend some reference to some of the older papers in this modeling approach discussed in B (Split-Fourier Method) which surely has some foundational reliance on the works of Knepp ("Multiple Phase-Screen Calculation of the Temporal Behavior of Stochastic Waves".

> **Proposed modification**: We thank the reviewer for the suggestion. The reference recommended by the reviewer has been added to the bibliography.

The graphs are inconsistent. For example, Fig. 2 and 7 both have a title above the graphs and the others don't. Several graphs (Figure 5, Figure 6) don't describe what the graduated bar on the right are (are they E(dB) as in other plots?)

> **Proposed modification**: The figures have been revised for consistency.

The application of Prony's method to a model for which Prony's method is fundamentally suited and which therefore yields such accurate results is not surprising. I think that the authors could have provided some cases to show the reader when the approach to break down. What are its limitations and how should one take that into account when using this method? It's the same approach that one takes in the comms/signal processing community when bit error rate graphs are shown - to expose the conditions under which the results are excellent and then the conditions under which the results are poor but explainable.

> **Response:** As noted in our response to comment B.V.3 from Reviewer 1, we have included an additional simulation scenario in Section IV.A to explicitly illustrate a limitation of the method. The results show that when the delay spread is short, the method is unable to reliably distinguish multiple paths.

Reviewer 6:

Comments to the Author
The paper employs Split-Step Fourier and Split-Step Wavelet methods to model and analyze multipath propagation in tropospheric communication channels. While the paper refines existing split-step simulation techniques, it does not introduce fundamentally new concepts or groundbreaking methods. The multipath extraction approach builds on established techniques rather than offering a novel paradigm. In addition, despite the wealth of studies that cover various methodologies, the reference list in the paper appears limited. This shortfall undermines the paper's ability to clearly position its contributions within the broader context of existing research.

Response: We thank the reviewer for the comment. The novelty of this article lies specifically in the multipath extraction from SSW simulations of long-range electromagnetic wave propagation in tropospheric environments, which, as the reviewer has noted, is not well covered in the literature. In response to the reviewer's concern about the limited reference list, we have substantially expanded the bibliography to include a broader range of studies. This improves the positioning of our work within the broader context of prior research and highlights its contribution.

Suggested References:

Reviewer: 1
Suggested References: N/A
Reviewer: 2
Suggested References: N/A
Reviewer: 3
Suggested References: N/A

Reviewer: 4
Suggested References: • R. Schmidt, "Multiple emitter location and signal parameter estimation," in IEEE Transactions on Antennas and Propagation, vol. 34, no. 3, pp. 276-280, March 1986, doi: 10.1109/TAP.1986.1143830.
→ This foundational paper introduces the MUSIC algorithm, a high-resolution method for multipath and direction-of-arrival estimation, which could be cited for comparison with Prony's method.
• Y. Hua and T. K. Sarkar, "Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise," in IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 38, no. 5, pp. 814-824, May 1990, doi: 10.1109/29.56027.
→ Offers an alternative to Prony's method that is more robust in noisy environments. Citing it would provide methodological depth
• T. Tunçkol, M. Biscarini and L. Luini, "Radiometric Estimation of Tropospheric Attenuation: A Mixed Physically Based/Machine Learning Approach," in IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1-13, 2024, Art no. 5301113, doi: 10.1109/TGRS.2024.3393506.
→Introduces a modern approach to modeling tropospheric attenuation through a hybrid physical/statistical technique that could complement the propagation scenarios considered in the manuscript.
• D. W. Matolak and R. Sun, "Air–Ground Channel Characterization for Unmanned Aircraft Systems—Part I: Methods, Measurements, and Models for Over-Water Settings," in IEEE Transactions on Vehicular Technology, vol. 66, no. 1, pp. 26-44, Jan. 2017, doi: 10.1109/TVT.2016.2530306
→ A key reference for realistic channel modeling in UAS applications, helping bridge simulation-based and empirical perspectives.

Reviewer: 5
Suggested References: D. L. Knepp, "Multiple phase-screen calculation of the temporal behavior of stochastic waves," in Proceedings of the IEEE, vol. 71, no. 6, pp. 722-737, June 1983, doi: 10.1109/PROC.1983.12660. keywords: {Stochastic processes; Radar scattering; Signal analysis; Partial differential equations; Time domain analysis; Satellite communication; Spaceborne radar; Frequency; Barium; Clouds},10.1109/PROC.1983.12660

Reviewer: 6
Suggested References: The paper does not sufficiently reference the extensive literature on tropospheric propagation and multipath modeling. Given the well-established research in this area, the limited citations weaken the paper's contextual grounding. A broader review of relevant studies would enhance its originality and strengthen its contribution.

Proposed modification: The references suggested by the reviewers have been incorporated into the bibliography.

An Extraction Method for the Multipath Characteristics of Simulated Tropospheric Propagation Channels

Hang Zhou and Alexandre Chabory

***Abstract*—Multipath propagation, resulting from signal reflections, diffractions, scatterings, and atmospheric ducting, poses significant challenges in wireless communication, navigation, and surveillance systems. This study focuses on evaluating the multipath characteristics in tropospheric propagation channels, particularly in air-ground systems. For long-range wave propagation in low-troposphere environments, split-step methods offer both precision and computational efficiency in modeling propagation at specific frequencies. We propose a multipath extraction approach to separate dominant propagation paths and associated multipath components by means of the post-processing of frequency-domain simulation results. The multipath extraction relies on the Prony's method, which allow to assess the multipath amplitude, phase, and delay from simulation results at a few frequencies. ~~We propose a multipath extraction approach to separate dominant propagation paths and associated multipath components from simulation results.~~ This research examines the multipath extraction method and validates the proposed models through numerical experiments across various scenarios and parameters. Simulation results are consistent with the two-ray model over flat ground. In this case, the extracted multipath reproduces the simulated channel with a root mean square error below –50 dB at all spatial points.**

***Index Terms*—multipath extraction, split-step wavelet, Prony's method, air-ground propagation channel, urban air mobility**

I. Introduction

Urban air mobility (UAM) will provide urban, suburban, and intra-urban air transport services, using vehicles that carry a limited number of passengers [1]. As UAM is integrated into existing airspace, challenges will arise, particularly in shared airspace with other types of vehicles. At high altitudes, the airspace will be shared with conventional piloted aircraft (airplanes and helicopters). At lower altitudes, the airspace will be shared with unmanned aircraft systems (UAS), which are increasingly used for many services, such as package delivery and ground surveillance.

The use of conventional air traffic control (ATC) communication-navigation-surveillance (CNS) systems for UAM at low altitudes is questionable [2], [3]. Indeed, the maximum capacity of some systems in densely populated areas is almost reached. This is notably the case for VHF communications and for automatic dependent surveillance–broadcast (ADS-B). For these systems, it will not be possible to integrate hundreds of new vehicles. Additionally, at low altitudes, the atmosphere, terrain, and obstacles such as buildings, bridges, towers, and generally large-size structures will inevitably partially block, reflect, and scatter the signal. These perturbations will be particularly significant when the radio link involves a ground station. [4]

Therefore, it is of primary importance to develop air-ground propagation channel models that are consistent with existing and future navigation and radar systems for unmanned aerial vehicle (UAV) and UAM. Extracting delays in the multipath is important in that case. For radar systems, paths whose delays differ significantly from the direct path yield spurious echoes. For navigation systems where the time of arrival or the time-difference of arrival are used, delayed multipath impairs the positioning accuracy. Besides, UAV and UAM operate in the very low troposphere, which is the domain where the environment (atmosphere and terrain) is most likely to produce multipath.

~~Therefore, it is of primary importance to develop air-ground propagation channel models that are consistent with existing and future UAM CNS systems. These models should be suitable for navigation systems based on ground beacons or ground-based augmentation system (GBAS), and surveillance systems such as radar or multilateration. This implies that the channel should be characterized in terms of several propagation paths, each with its own attenuation and delay.~~

In moderately populated areas, particularly in the context of intra-urban air transport, the system range can extend beyond several tens of kilometers. In such situations, both the terrain and the atmosphere affect the propagation channels in terms of attenuation and delay. In this case, two-dimensional methods are suitable for modeling wave propagation.

The propagation channel model can be based either on measurements or simulations. Models based on measurements require extensive measurement campaigns, using, for example, a channel sounding system [5]–[7]. The limitations of this approach are their cost and the fact that they are site-specific, i.e., the developed model will only be consistent with situations that are similar to the initial measurement environment. To assess the multipath parameters from measurements, superresolution algorithms are generally used [8]–[12]. These assume that the received signal is a sum of independent signals in the presence of white noise. This works well for channel soundings conducted with a large enough frequency band and a sufficient number of samples.

Alternatively, computational electromagnetics offer several tools to simulate propagation channels [13], [14]. If the

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environment is known, these tools can be applied to any site where a ground antenna is intended to be placed. However, the computational cost of these tools can be significant. Additionally, many computational electromagnetic tools work in the frequency domain, which means that propagation delays are not directly obtained. Therefore, there is a need to assess multipath from electromagnetic simulations in the frequency domain. The simulation should be performed at a few frequencies for the computation load to remain acceptable.

In the frequency domain, ray methods can be used for predicting atmospheric propagation. Delays of multipath are then simply related to the electrical length of the ray paths. However, these methods have limitations in certain conditions, especially in atmospheric ducting. When atmospheric refraction causes rays to bend and overlap, calculation errors could occur. [15] The split-step Fourier method (SSF) is another computational electromagnetics tool that works in the frequency domain and it is often used in low tropospheric environments. NASA's Urban Air Mobility Concept of Operations [16] envisions UAM aircraft operating at cruising altitudes generally between 1,200 and 5,000 feet above ground level (AGL). These altitude ranges imply that UAM primarily operates at relatively low elevation angles when communicating with ground-based CNS systems. As a result, they are more vulnerable to multipath propagation effects caused by atmospheric ducting, surface reflections, and terrain irregularities. In contrast, for near-zenith propagation paths, the risk of multipath delays is small. SSF employs fast Fourier transforms (FFT) to speed up wave transformations between spatial and spectral domains over distances [13], [17]. To enhance the computational efficiency, the split-step wavelet (SSW) method replaces FFT with fast wavelet transform (FWT), iteratively transitioning between spatial and wavelet representations [14]. SSW reduces computational burden while maintaining similar modeling capabilities for inhomogeneous refractivity, ground boundary conditions, and irregular terrain [18], [19].

To extract multipath components, signal processing techniques can be broadly categorized into three classes. Parametric methods, such as Prony's method [20], [21], the Matrix Pencil method [22], MUSIC, and ESPRIT [23], model the received signal as a sum of complex exponentials. These methods offer high-resolution estimation of multipath parameters, including path delays and amplitudes. However, techniques like MUSIC and ESPRIT typically require a large number of frequency samples and are optimized for scenarios with additive noise. Time-domain inversion techniques aim to reconstruct impulse responses or estimate multipath parameters directly from observed signals. Representative approaches include deconvolution, sparse recovery, matching pursuit, and CLEAN-based algorithms [24], [25]. In particular, IFFT-based reconstruction provides an efficient way to convert frequency-domain data into approximate time-domain responses. While these methods are computationally efficient and easily interpretable, their resolution is fundamentally constrained by the signal bandwidth and frequency sampling density. Machine learning–based methods, including convolutional neural networks (CNNs), graph-based models, and transformers, have also been applied to extract propagation parameters from electromagnetic simulation data [26]–[28]. These data-driven approaches exhibit strong generalization capabilities across complex environments. However, they typically require large volumes of labeled training data and may lack physical interpretability. Overall, these three categories of techniques offer complementary strengths for analyzing and extracting multipath effects in electromagnetic wave propagation, each suited to different application contexts and data characteristics.

The objective of this article is to assess the multipath characteristics of the low-troposphere propagation channel. The novelty of this work lies in combining split-step methods for computing the fields at a small number of frequencies with an efficient method for extracting multipath components in terms of amplitude, phase, and delay. Several numerical experiments are conducted, and the flat-ground scenario is validated against the two-ray model.

~~The objective of this article is to use split-step methods for computing multipath propagation channel in the low troposphere. A method is proposed to extract the multipath characteristics in terms of amplitude, phase, and delay from simulation results at a few frequencies. Several numerical experiments are performed. The one over planar ground is validated against the two-rays model.~~

The structure of the article is as follows: Section II outlines the simulation methodologies, detailing the split-step Fourier and split-step wavelet methods. Section III delves into the multipath extraction techniques. Section IV presents numerical simulations conducted across various configurations to evaluate the proposed multipath extraction methods.

II. Presentation of Simulation Methods

In this section, we first present the propagation configuration and then introduce two split-step simulation methods for calculating wave propagation in the low troposphere: the split-step Fourier method and the split-step wavelet method.

A. Configuration

The objective is to simulate the propagation in a vertical plane (x, z) and y is the direction of invariance. As plotted in Fig. 1, the source is located at $x < 0$, and the electromagnetic field is known at $x = 0$. For a wave mainly propagating along the $+x$ direction, a field with reduced phase variation, denoted as $u(x, z)$, is used. The propagation is computed in the region $x > 0$ and $z \geq 0$. We assume a time dependence of the form $e^{j\omega t}$ for the fields, where ω is the angular frequency. Any field can be decomposed into the transverse electric (TE) and transverse magnetic (TM) components with respect to the z-direction. For simplicity, we only present the TE component. The study of the TM component follows a similar approach.

For numerical purposes, the computational domain is discretized and finite in size. The vertical domain is bounded by $z \in [0, z_{\max})$ and the horizontal range is limited to $x \in [0, x_{\max})$. The discretization steps are Δx and Δz, and the number of points in two direction are given by $N_x = x_{\max}/\Delta x$ and $N_z = z_{\max}/\Delta z$. The following uniform grid is used for the discretization

$$\begin{aligned} x &= p_x \Delta x \quad \text{for} \quad p_x = \{0, \ldots, N_x - 1\}, \\ z &= p_z \Delta z \quad \text{for} \quad p_z = \{0, \ldots, N_z - 1\}. \end{aligned} \tag{1}$$

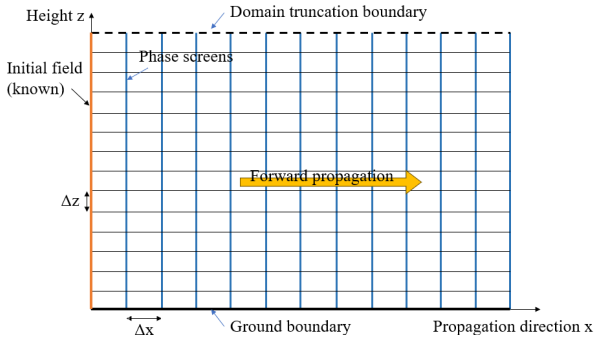


Fig. 1. Configuration of electromagnetic wave propagation simulation.

The top boundary is set as the truncation boundary of the domain. On that boundary, the Sommerfeld radiation condition should be satisfied. The ground boundary is modeled as either a perfect electric conductor (PEC) or an impedance boundary, with an irregular relief considered.

An inhomogeneous atmosphere is assumed, with the atmospheric characteristics represented by the refractive index n. Since n is close to 1, it is convenient to use the refractivity N, defined by $N = 10^6 \times (n-1)$.

B. Split-step Fourier method

The Parabolic Equation (PE) approximates the wave equation along a paraxial direction and is implemented using either finite differences (FD) or SSF. FD is particularly advantageous for addressing complex boundary conditions, while SSF is more computationally efficient, especially for long-range propagation scenarios. The SSF method achieves this efficiency by alternately transforming the wave field between spatial and spectral domains using FFT. In low-tropospheric wave propagation, SSF includes the modeling of atmospheric refractivity, ground characteristics, and terrain relief. To do so, phase-screen methods spatially account for refractivity variations, the discrete mixed Fourier transform accurately handles impedance ground reflections in the spectral domain, and specialized algorithms are employed to integrate terrain effects [13]. The simulation starts at x = 0 where the field is assumed to be known. This field is generally obtained either from the transmitting antenna pattern, or from an equivalent radiating aperture [13]. The propagation from x to $x + \Delta x$ is computed step by step as follows: ~~The simulation starts at $x = 0$ and proceeds by computing the propagation from x to $x + \Delta x$ in the subsequent steps:~~

1) The field $u(x,z)$ undergoes transformation into the spectral domain, resulting in a vector $U(x)$ through FFT (referred to as F).
2) Free-space propagation over Δx is performed in the spectral domain by multiplying $U(x)$ by a propagator P. Thus, $U_{\text{fs}}(x+\Delta x) = PU(x)$. The wide-angle parabolic equation is applied with the range of applicable angles is approximately –45° to +45°.
3) The spatial representation of the propagated field $u_{\text{fs}}(x+\Delta x, z)$ is obtained by applying the inverse FFT (denoted by F^{-1}) on $U_{\text{fs}}(x+\Delta x)$.
4) Methods for apodization, atmosphere, and relief effects [13] are applied in the spatial domain, represented by an operator D, resulting in $u(x+\Delta x, z) = Du_{\text{fs}}(x+\Delta x, z)$.

Consequently, the propagation from x to $x+\Delta x$ is simulated step by step using

$$u(x+\Delta x, z) = DF^{-1}PFu(x,z). \tag{2}$$

These steps are iterated until the maximum computation distance $x_{\max}$ is reached.

C. Split-step wavelet method

To enhance the computational efficiency, we have introduced a wavelet-based technique called the split-step wavelet (SSW) method [14]. Subsequently, SSW has been improved in terms of memory usage and versatility by utilizing the propagation of elementary wavelets [5]. SSW employs FWT instead of FFT. The computational efficiency of FWT is significantly lower than that of FFT. SSW iteratively models electromagnetic wave propagation over increasing distances by transitioning between spatial and wavelet representations. The method proceeds as follows:

1) The field $u(x,z)$ is transformed into a sparse wavelet vector $\tilde{U}(x)$ using FWT (denoted as W) and compression (denoted as C), expressed as $\tilde{U}(x) = CWu(x,z)$.
2) Free-space propagation is performed in the wavelet domain. The coefficients vector $\tilde{U}_{\text{fs}}(x+\Delta x)$ after free-space propagation over Δx is computed using a pre-calculated free-space propagation matrix M. This matrix models wavelet-to-wavelet propagation, yielding $\tilde{U}_{\text{fs}}(x+\Delta x) = M\tilde{U}(x)$.
3) The field $u_{\text{fs}}(x+\Delta x, z)$ propagated in free space is reconstructed by the inverse FWT (W^{-1}) from the propagated wavelets: $u_{\text{fs}}(x+\Delta x, z) = W^{-1}\tilde{U}_{\text{fs}}(x+\Delta x)$.
4) Apodization, atmospheric, and relief effects, similar to those in SSF, are then applied in the spatial domain using the same operator D.

In SSW, the propagation from x to $x+\Delta x$ is simulated step by step using

$$u(x+\Delta x, z) = DW^{-1}MCWu(x,z), \tag{3}$$

and these steps are iterated until $x_{\max}$.

Previous studies [14], [19] have shown that SSW maintains the accuracy of SSF with reduced computation time. For this reason, in this article, only simulations with SSW are considered.

III. Multipath Extraction Method

This section presents the method employed for modeling multipath signals and extracting their parameters. The accuracy of the extraction method is then analyzed. Finally, a post-processing method is proposed to group multipath components with similar delays.

A. Signal modeling

The received signal in a complex propagation environment with multiple paths can be expressed as the superposition of delayed and attenuated echoes:

$$u(f) = \sum_{p=0}^{n_\mathrm{p}-1} a_p e^{-j2\pi f \tau_p}, \tag{4}$$

where f represents the electromagnetic wave frequency, n_p is the total number of propagation paths, and a_p and τ_p denote the complex amplitude and delay of the p-th path, respectively. This equation represents a wideband channel model, as indicated in [8]. In this study, no noise is considered, as the aim is to efficiently extract the multipath components in terms of amplitude, phase, and delay from simulation results using only a few frequency samples. The numerical errors inherent in the simulation cannot be appropriately modeled as additional Gaussian noise.

A limitation of all parametric multipath estimation methods (Prony's method, ESPRIT, MUSIC, etc.) is that the number of propagation paths n_p must be specified in advance. However, this value is generally not known a priori and cannot be reliably determined without prior knowledge of the channel structure. In typical ground-based line-of-sight propagation scenarios, the number of dominant multipath components is generally low due to the limited number of significant reflectors near the ground. Based on prior studies, we expect that a small number of paths (usually fewer than 5) are sufficient to capture the essential multipath structure.

The propagation simulation methods such as SSF and SSW are utilized to compute $u(f_n)$ at multiple regularly spaced frequencies $f_n = f + n\delta f$, where $n \in \{0, \cdots, n_\mathrm{f} - 1\}$ and n_f is the total number of frequency samples. For simplicity, let $u_n = u(f_n)$.

The objective is to determine the multipath parameters a_p and τ_p for $p \in \{0, \cdots, n_\mathrm{p} - 1\}$ from u_n, using as few frequency samples n_f as possible. After sampling, the problem can be reformulated as

$$u_n = \sum_{p=0}^{n_\mathrm{p}-1} a_p {z_p}^n, \qquad \text{with } z_p = e^{-j2\pi\delta f \tau_p}, \tag{5}$$

where $n = 0, \cdots, n_\mathrm{f} - 1$. The right-hand-side of (5) can be cast as matrix-vector product, where the matrix elements are

$$\mathbf{M}_{\tau_0,\cdots,\tau_{n_\mathrm{p}-1}} = \begin{bmatrix} 1 & \cdots & 1 \\ z_0 & \cdots & z_{n_\mathrm{p}-1} \\ \vdots & & \vdots \\ {z_0}^{n_\mathrm{f}-1} & \cdots & (z_{n_\mathrm{p}-1})^{n_\mathrm{f}-1} \end{bmatrix}, \tag{6}$$

and the vector components are

$$\mathbf{u} = \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n_\mathrm{f}-1} \end{bmatrix}, \qquad \mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n_\mathrm{p}-1} \end{bmatrix}. \tag{7}$$

Thus, (5) can be expressed in matrix form as follows

$$\mathbf{u} = \mathbf{M}_{\tau_0,\cdots,\tau_{n_\mathrm{p}-1}} \mathbf{a}, \tag{8}$$

For simplicity, the indices $\tau_0, \cdots, \tau_{n_\mathrm{p}-1}$ are omitted from $\mathbf{M}$. The matrix $\mathbf{M}$ exhibits a structure known as a Vandermonde matrix.

B. Prony's method

Prony's method [20], [21] is based on the principle that any series in the form of (5) satisfies a homogeneous difference equation. This implies the existence of coefficients $h_1, \cdots, h_{n_\mathrm{p}}$ such that

$$u_n = h_1 u_{n-1} + h_2 u_{n-2} \ + \cdots + h_{n_\mathrm{p}} u_{n-n_\mathrm{p}} \tag{9}$$

In other words, u_n can be interpreted as the output of a finite impulse response (FIR) filter. The corresponding characteristic polynomial is given by

$$P = \sum_{p=0}^{n_\mathrm{p}} h_p z^{p-n_\mathrm{p}}, \qquad \text{with} \quad h_0 = -1. \tag{10}$$

This polynomial is related to the Z-transform of the filter. The roots of this polynomial are z_p. From (4), the multipath delays derive from the phases of these roots.

For $n_\mathrm{f} \geq 2n_\mathrm{p}$, the following steps estimate τ_p and a_p from u_n:

- Calculate h_p:
  Using (9), h_p can be obtained by solving the linear system

$$\begin{bmatrix} u_{n_\mathrm{p}-1} & \cdots & u_0 \\ u_{n_\mathrm{p}} & \cdots & u_1 \\ \vdots & & \vdots \\ u_{n_\mathrm{f}-2} & \cdots & u_{n_\mathrm{f}-n_\mathrm{p}-1} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_{n_\mathrm{p}} \end{bmatrix} = \begin{bmatrix} u_{n_\mathrm{p}} \\ \vdots \\ u_{n_\mathrm{f}-1} \end{bmatrix}. \tag{11}$$

  If $n_\mathrm{f} = 2n_\mathrm{p}$, the system is square, and the solution fits the data exactly. For $n_\mathrm{f} > 2n_\mathrm{p}$, the system is overdetermined and solved using the least squares method.
- Compute the delays τ_p:
  The roots z_p of the polynomial (10) are calculated, from which the delays are obtained as

$$\tau_p = -\frac{\angle z_p}{2\pi\delta f} \tag{12}$$

- Solve for $\mathbf{a}$:
  Using the values of z_p in $\mathbf{M}$, the linear system (8) is solved to determine $\mathbf{a}$. Here, the linear system is of size $n_f \times n_p$. It is always overdetermined since (11) imposes $n_f \geq 2n_p$.

For the special case of a single propagation path ($n_\mathrm{p} = 1$) with two frequency samples, the polynomial reduces to first-order, with the root

$$z_0 = \frac{u_1}{u_0} \qquad \Rightarrow \qquad \tau_0 = -\frac{\angle(u_1/u_0)}{2\pi\delta f}. \tag{13}$$

Since the group delay of a signal is defined by -$d\angle u/d\omega$, the Prony's method provides in this case the finite-difference approximation of the group delay. The phase ambiguity conditions are further discussed in [29].

C. Accuracy

Because of the possible phase ambiguity, the estimated phase of z_p in (12) will be $2\pi\delta f\tau_p+2l\pi$, where l is an integer. This implies that τ_p can be directly extracted, i.e., without phase ambiguity from $\angle z_p$ if the condition $2\pi\delta f\tau_p \leq 2\pi$ is satisfied. Hence, the following criteria must be respected

$$\delta f \leq 1/\tau_p. \tag{14}$$

However, a small value of δf implies that the phase in z_p will also be very small, which may be impaired by the simulation's accuracy. Additionally, for small values of δf, the linear system that must be solved in Prony's method will be very poorly conditioned since successive values of u_n will be very similar.

D. Reduction of the number of multipath

The application of Prony's method may yield results containing very weak multipath components or multiple paths with nearly identical delays. To enhance the interpretability of the simulation results, we propose a post-processing step to simplify the multipath representation.

First, multipath components with amplitudes below a predefined threshold a_{thres} are suppressed. The modified amplitudes are given by

$$a_p^{\text{thres}} = \begin{cases} a_p & \text{if } |a_p| > a_{\text{thres}} \\ 0 & \text{if } |a_p| \leq a_{\text{thres}}. \end{cases} \tag{15}$$

Due to various uncertainties (e.g., atmospheric refraction index, terrain relief, ground composition) and simulation approximations, the channel response is only approximately modeled. Therefore, in numerical experiments, the threshold below which paths are removed is set to -60 dB.

Second, multipath components with delays differing by less than a predefined threshold τ_{min} are grouped into a single equivalent path. Consider two multipath components, indexed as p and p', with delays τ_p and $\tau_{p'}$, respectively. If their delay difference satisfies $|\tau_p - \tau_{p'}| < \tau_{\text{min}}$, the components are merged into a single equivalent path with the following parameters

$$a^{\text{clustered}} = a_p + a_{p'}, \qquad \tau^{\text{clustered}} = \frac{|a_p|^2\tau_p + |a_{p'}|^2\tau_{p'}}{|a_p|^2 + |a_{p'}|^2}. \tag{16}$$

The choice of τ_{min} depends on the parameters of the system under study, notably its bandwidth. Moreover, for radar and navigation systems, the delay resolution typically corresponds to a path length of at least 1 m. Therefore, τ_{min} is set to $1/c_0$, where c_0 denotes the speed of light in vacuum. [30], [31]

E. Complexity analysis

The total computational complexity of the SSW method combined with the multipath extraction method, denoted as $\mathcal{C}_{\text{TC}}$, can be expressed as

$$\mathcal{C}_{\text{TC}} = n_f\mathcal{C}_{\text{SSW}} + n_x n_z \mathcal{C}_{\text{ext}}, \tag{17}$$

where

- $\mathcal{C}_{\text{SSW}}$ denotes the computational cost of the SSW method. As detailed in [32], the complexity of the fast wavelet transform and the associated compression operations is approximately $\mathcal{O}(N_x N_z)$.
- $\mathcal{C}_{\text{ext}}$ represents the complexity of the multipath extraction at a single point. This includes solving (11), computing the roots of a polynomial of degree n_p, and solving (8), a linear system of size n_f.

In the numerical experiments, only a limited number of frequencies n_f, and hence a small number of propagation paths n_p, are extracted, the overall computational complexity is generally dominated by $n_f\mathcal{C}_{\text{SSW}}$.

IV. Numerical experiments

This section explores numerical simulations using various configurations of terrain, atmospheric conditions, and simulation parameters. Prony's method described in subsection III-B and the post-processing method introduced in subsection III-D are employed to efficiently extract multipath components from the simulation results using SSW.

A. Wave propagation over a planar PEC ground

The initial simulation explores the propagation over a planar perfect electric conductor (PEC) ground in a homogeneous atmosphere. A complex source point (CSP) [33] serves as the wave's source, located at $(x_{\text{s}}, y_{\text{s}}, z_{\text{s}})$ with $x_{\text{s}} \in \mathbf{C}$ such that $x_{\text{s}} = x_{w0} - jkW_0^2/2$, where k is the wavenumber and W_0 is the source's width. In free space, the wave radiation can be analytically derived using the Green's function and its expression in 2D is given by

$$u(r) = A\frac{j}{4}H_0^{(2)}(kr), \tag{18}$$

where A is a normalization constant, $H_0^{(2)}$ is the Hankel function of the second kind, and

$$r = \sqrt{(x - x_{\text{s}})^2 + (z - z_{\text{s}})^2}, \tag{19}$$

with the condition that the real part of r is positive. CSP represents an antenna whose radiation pattern is approximately Gaussian. This could correspond to a Potter or corrugated horn antenna. Moreover, several methods exist to expand any radiation into a linear combination of Gaussian waveforms. Therefore, by extension, the method is applicable to any electromagnetic waves, including those characterized by an arbitrary CNS antenna radiation pattern.

In the presence of a PEC planar ground, an image CSP is introduced. The exact solution is obtained by superposing the fields radiated from the original CSP and the field from the image CSP, the latter of which is multiplied by -1. Two propagation paths can be defined, one corresponding to each CSP. The parameters of the wave source are listed in Table I.

The simulation parameters are detailed in Table II. In SSW, an absorbing layer is placed at the upper boundary of the computation domain to prevent reflection. This layer occupies a fixed 30% of the domain. In Table II, $z_{\text{max_cd}}$ denotes the maximum altitude of the effective computational domain,

TABLE I
PARAMETERS OF THE COMPLEX SOURCE POINT

Parameters	Values
EM wave frequency f	5 GHz
Polarization	TE
W_0	0.5 m
x_{w0}	-100 m
y_s	0 m
z_s	200 m

TABLE II
SIMULATION PARAMETERS USED FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION A.

Parameters	Descriptions	Values
$z_{\max}$	Maximum altitude of the total simulation domain	1000 m
$z_{\max_cd}$	Maximum altitude of the computation domain	700 m
Δ_z	Vertical discretization step	0.2 m
N_z	Number of vertical grid points	5000
$x_{\max}$	Maximum horizontal range of the computation domain	15 km
Δ_x	Horizontal discretization step	30 m
N_x	Number of horizontal grid points	500

excluding the absorbing layer. The compression threshold in SSW is set to 10^{-8}, which corresponds to a maximum compression error of -100 dB, as described in [14], [34]. ~~The SSW method is configured to maintain a maximum compression error of -100 dB, as described in~~ [14].

Prony's method is used to extract multipath components with $n_{\mathrm{p}} = 2$. To capture 2 distinct multipaths, it is necessary to propagate at least $n_{\mathrm{f}} = 4$ different frequencies, uniformly spaced by δf. Here, we always set $n_f = 2n_p$ in the simulations. According to (14), δf is set to 0.5 MHz, so that multipath with delays up to 600 m can be extracted without ambiguity. This results in a sequence of frequencies: 5.0000, 5.0005, 5.0010, and 5.0015 GHz. In the post-processing, $a_{\mathrm{thres}} = 10^{-3}$ and $\tau_{\min} c_0 = 1$ m are set, as introduced in subsection III-D.

The normalized propagated field is illustrated in Fig. 2. The upper zone above the dashed line corresponds to the apodization/absorbing layer. In this simulation, this zone is shown for illustration purposes only and is omitted in the results of the other simulations for simplicity. The direct and reflected waves superimpose to form interference fringes, each with different attenuation and delay. After applying the multipath extraction method, the first and second extracted path components are presented in Fig. 3. The left subplots display the group delays of the reduced field, while the right subplots depict the amplitudes corresponding to these components. To simplify interpretations of the simulations results, delays are characterized as the distance τc_0. The top row highlights the first path component, which is the dominant one. It corresponds to direct propagation, representing the wave radiated directly from the source. The second row shows the reflected wave. The grazing reflection is categorized as part of the direct propagation path rather than being grouped with the reflected waves in the second figure. This is because very near the ground and at grazing incidence, the direct and the

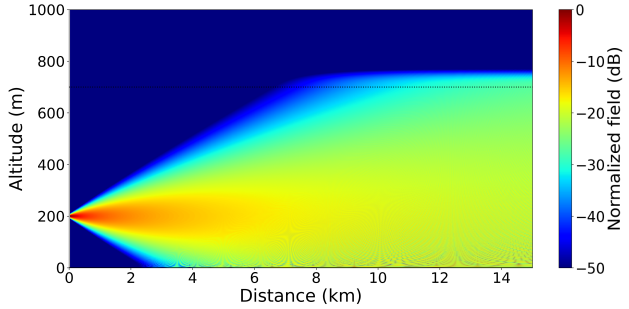


Fig. 2. Normalized electromagnetic field computed using SSW over a planar PEC ground in a homogeneous atmosphere (Source height: 200 m).

reflected waves travel almost the same distance, that is to say, the rays are almost of the same length. Thus, the algorithm cannot separate the direct field from the ground reflection near the ground. The delays and amplitude of the reflected wave obtained using SSW, followed by the extraction method, are compared with the two-ray model, which includes both line-of-sight and ground reflection. The results for the final vertical (at the maximum range 15 km) are presented in Fig. 4a and Fig. 4b. The extraction results align with the theoretical analysis.

To validate the multipath extraction results, we evaluated the cumulative error between the reconstructed fields (obtained using (4)) and the simulated fields obtained with SSW at n_{f} frequencies. This error is quantified using the root mean square (RMS) error in dB

$$\epsilon_{\mathrm{RMS}} = 10\log \frac{\sum_{n=0}^{n_{\mathrm{f}}-1} \left| u_n - \sum_{p=0}^{n_{\mathrm{p}}-1} a_p e^{-2j\pi n \delta f \tau_p} \right|^2}{\sum_{n=0}^{n_{\mathrm{f}}-1} |u_n|^2}. \tag{20}$$

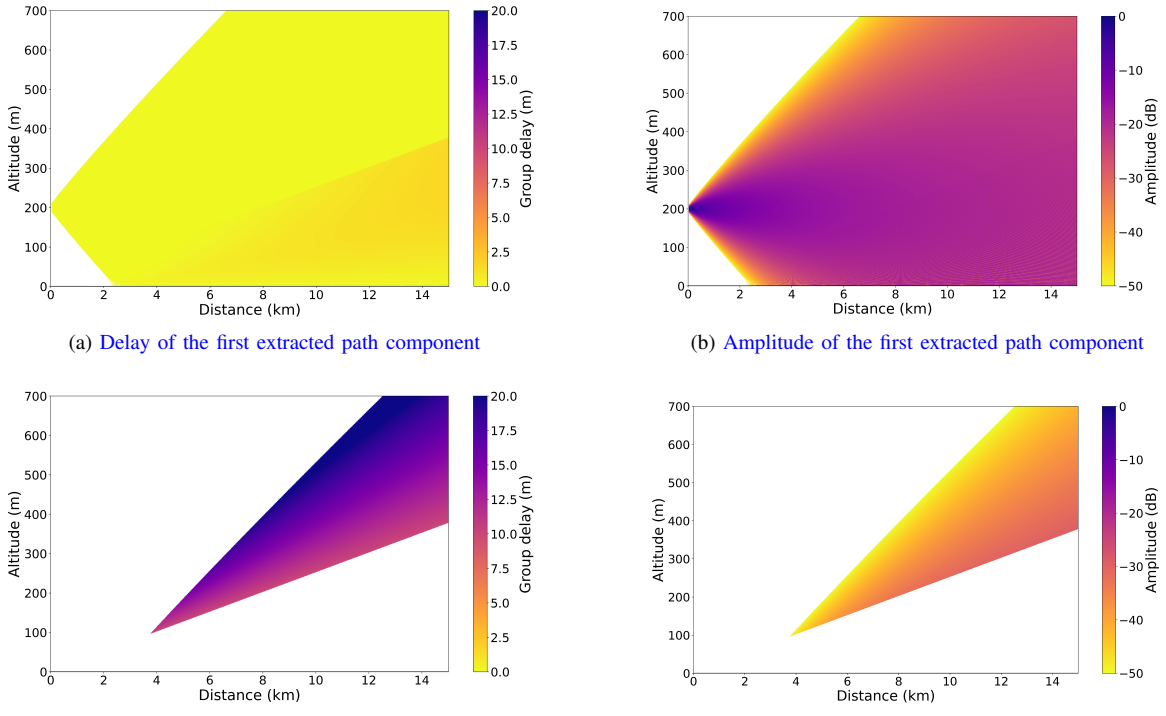
The computed error values are plotted in Fig. 5, showing a difference of less than -50 dB within the paraxial range from the CSP. This low error magnitude confirms the validity of the proposed multipath extraction method.

Additionally, we compare the simulation result $u(f)$ at a frequency of $f = 5000.75$ MHz with the reconstructed field obtained from using the multipath extraction method. The difference between these two fields is

$$\epsilon^{'}_{\mathrm{RMS}} = 10\log \frac{\left| u(f) - \sum_{p=0}^{n_{\mathrm{p}}-1} a_p e^{-2j\pi f \tau_p} \right|^2}{|u(f)|^2}, \tag{21}$$

where the summation represents the reconstructed field composed of the multipath components. The error is plotted in Fig. 6, showing that the maximum difference between the simulation and the reconstructed field is -39.7 dB. This small difference indicates a high accuracy of the multipath extraction method.

When the source height is set to 20 m, the delay between the reflected and direct paths is less than 2 m. The normalized



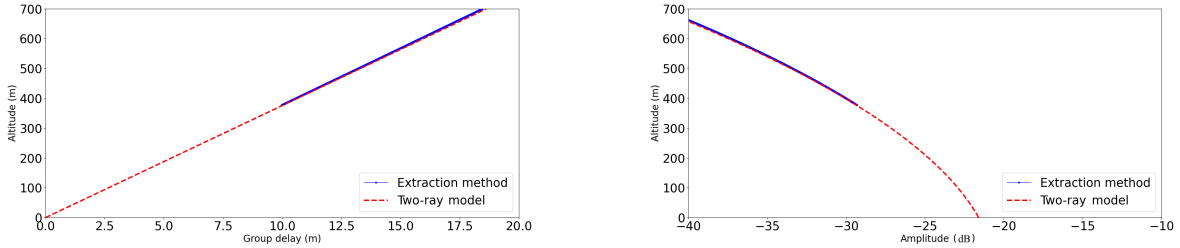
(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

Fig. 3. Extracted multipath components in propagation over a planar PEC ground in a homogeneous atmosphere (Source height: 200 m).



(a) Delay of the second path component: extraction method compared to analytical solution

(b) Altitude of the second path component: extraction method compared to analytical solution

Fig. 4. Comparison of the delay and altitude of the second path component between the extraction method and the analytical solution at the maximum range (15 km) over a planar PEC ground in a homogeneous atmosphere.

propagated field is illustrated in Fig. 7. After applying the multipath extraction method, the first and second components are presented in Fig. 8. The amplitude of the second component is negligible (under -50 dB). In this case, only one dominant path is sufficient to characterize the channel. This shows a limitation of the method: it can effectively separate different components only when there is a sufficient delay between them.

B. Wave propagation over an irregular dielectric ground

The second simulation examines the propagation characteristics of a CSP over an irregular dielectric ground. The ground profile, depicted as the black region in Fig. 9, represents a realistic terrain with a maximum height of approximately 100 m. We adopt a terrain generation method in which the terrain

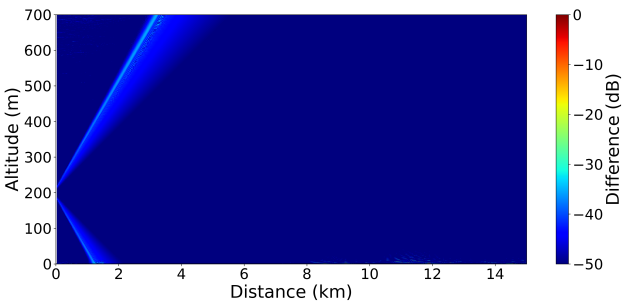


Fig. 5. Root mean square error between the reconstructed field using multipath extraction and the simulated field from SSW across n_f frequencies.

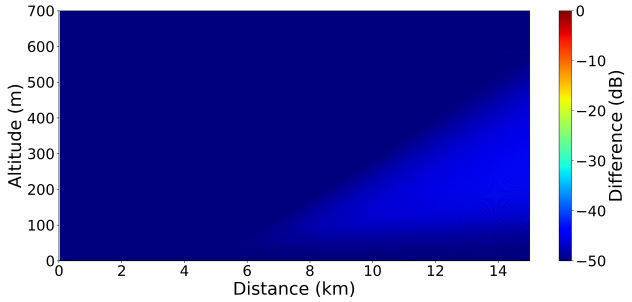


Fig. 6. Root mean square error between the reconstructed field using multipath extraction and the simulated field from SSW at the frequency 5000.75 MHz.

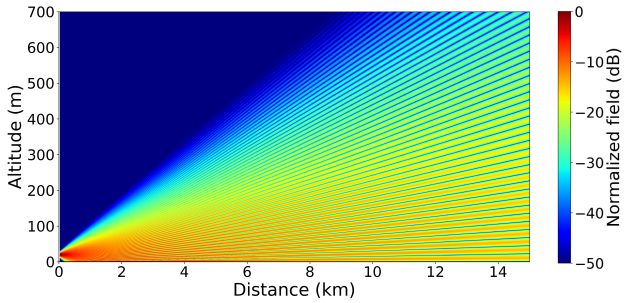


Fig. 7. Normalized electromagnetic field computed using SSW over a planar PEC ground in a homogeneous atmosphere (Source height: 20 m).

profile is formed by the superposition of random surfaces defined across multiple scales [35]. At each scale, random height values are sampled and interpolated, typically using linear or cosine interpolation, to create smooth transitions. The resulting multi-scale layers are then weighted and summed to produce a one-dimensional terrain that exhibits both randomness and natural continuity. The ground's relative permittivity is set to $\epsilon_{\mathrm{r}} = 15$, and its conductivity is $\sigma = 0.5$ S/m.

The simulation considers $n_{\mathrm{f}} = 8$ discrete frequencies, uniformly spaced by $\delta f = 5$ MHz. The normalized propagated field, calculated using SSW, is presented in Fig. 9. The extracted propagation paths, limited to $n_{\mathrm{p}} = 4$, are shown in Fig. 10. The first component in Fig. 10 corresponds to the direct wave. The next components represent multipath effects, which occur when ground reflections and diffractions generate fields that propagate in directions significantly deviating from the field emitted by the CSP.

C. Wave propagation in a complex environment

The propagation of a CSP within a complex environment is analyzed, focusing on long-range wave propagation over an irregular dielectric ground under a ducting atmosphere with accounting the Earth's curvature. The atmospheric model, illustrated in Fig. 11, is defined by a trilinear profile commonly used to represent ducting layers in the atmosphere [13]. The model is characterized by the following parameters: $z_{\mathrm{b}} = 200$ m and $z_{\mathrm{t}} = 1000$ m, which denote the bottom and top heights of the ducting layer, respectively; and $c_1 = -0.1$ M-unit/m, $c_2 = 0.118$ M-unit/m, which represent the vertical gradients of modified refractivity within different sublayers. In this article, the Earth curvature is realized by using modified refraction index, as introduced in [13]. ~~The atmospheric model, shown in Fig. 11, is defined by the following parameters: $z_{\mathrm{b}} = 200$ m, $z_{\mathrm{t}} = 1000$ m, $c_1 = -0.1$ M-unit/m, and $c_2 = 0.118$ M-unit/m.~~

Details of the source configuration and simulation parameters are summarized in Table III. Simulations are performed at 8 frequencies with a frequency increment of $\delta f = 1.5$ MHz. Fig. 12 illustrates the normalized electromagnetic wave fields computed using the SSW method, while Fig. 13 presents the extracted propagation components derived from the simulations.

The electromagnetic wave propagation is significantly influenced by the refractive index gradient within the atmospheric duct. Above 200 m, waves bend upward due to the decreasing refractive index, whereas below 200 m, they are guided downward by the ducting effect. Upon interacting with the irregular terrain, reflection and diffraction phenomena produce complex multipath effects, as depicted in Fig. 12.

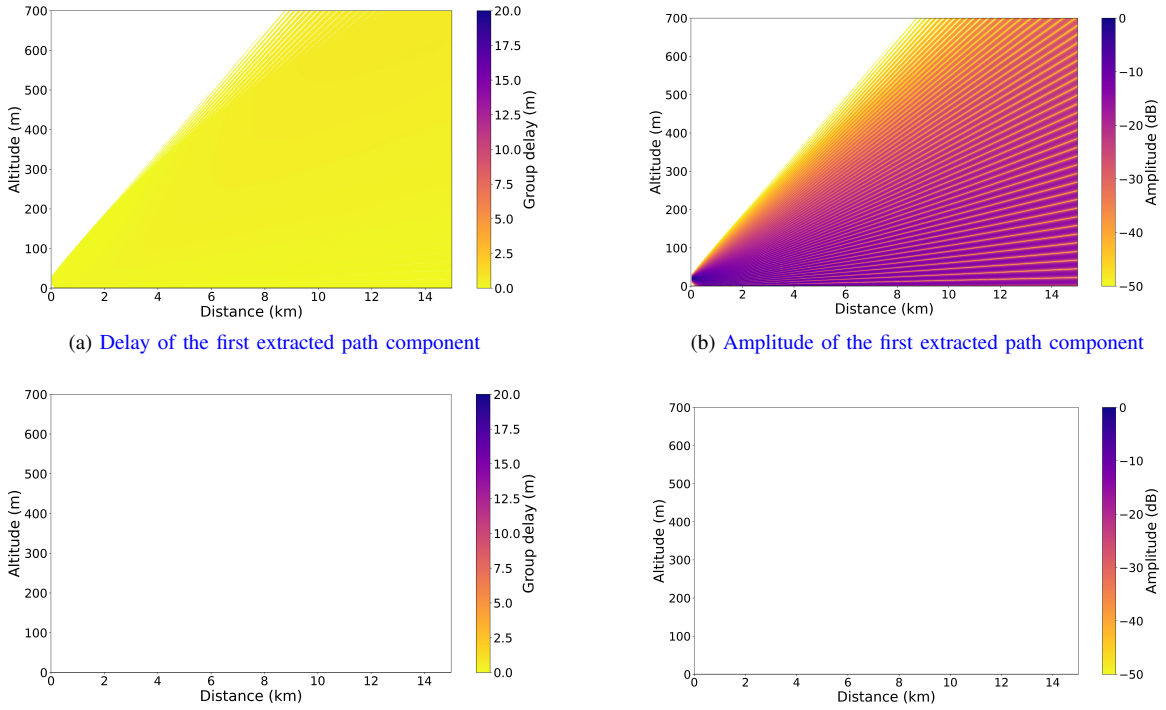
After applying the multipath extraction method, four path components are identified and shown in Fig. 13. The first component, represented in the top row, predominantly includes direct waves bent by the refractive index gradient within the duct. The second component mainly consists of waves reflected or diffracted by the irregular ground. The third and fourth components correspond to waves with substantial group delays, arising from interactions between direct waves and terrain-reflected waves. This classification highlights the intricate wave behavior within a ducting atmosphere.

TABLE III
SOURCE AND SIMULATION PARAMETERS FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION C.

Parameters	Values	Parameters	Values
f	2 GHz	$z_{\max}$	2000 m
Polarization	TE	$z_{\max_cd}$	1400 m
W_0	5 m	Δ_z	0.2 m
x_{w0}	-100 m	N_z	10000
y_s	0 m	$x_{\max}$	250 km
z_s	200 m	Δ_x	500 m
n_{f}	8	N_x	500
δ_{f}	1.5 MHz	n_{p}	4

D. Propagation with high obstacles

A simulation considering several large high-rise buildings is conducted, which can be modeled using vertical screens [36]. The urban environment is modeled using a randomly generated building-like terrain profile. A total of $N_{\mathrm{bu}} = 8$ buildings are randomly distributed along the horizontal axis, with each building's position independently selected. As the typical width of high-rise buildings (approximately 30-100 m) is much smaller than the simulation grid spacing ($\Delta x = 500$ m), buildings are approximated as vertical screens. Building heights are randomly sampled from the interval $[0, z_{\max_\mathrm{bu}}]$, with $z_{\max_\mathrm{bu}} = 400$ m. The distribution of vertical screens is represented by the black line in Fig. 14. The simulation incorporates the standard atmospheric model while accounting for the Earth's



(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

Fig. 8. Extracted multipath components in propagation over a planar PEC ground in a homogeneous atmosphere (Source height: 20 m).

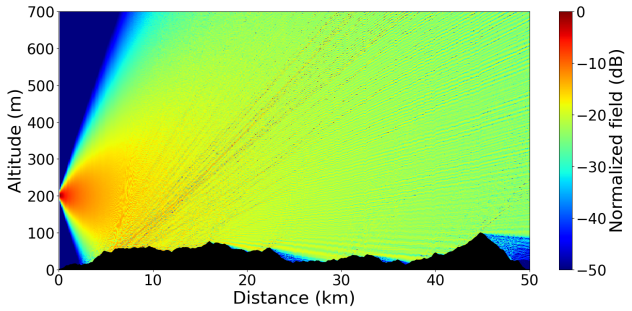


Fig. 9. Normalized electromagnetic wave field with SSW in propagation over an irregular dielectric ground in a homogeneous atmosphere.

curvature. Details of the source configuration and simulation parameters are provided in Table IV.

The simulation results, displayed in Fig. 14 and Fig. 15, reveal distinct propagation behaviors. Consistent with the previous simulations, the first component is predominantly associated with direct wave propagation, as the electromagnetic waves travel along a direct path unobstructed by urban structures. The second component corresponds to waves influenced by reflections from the ground surface. For the second component, in addition to ground reflections, there are also components of waves diffracted by the screen, whose propagation directions are nearly parallel to the ground. The third and fourth components capture reflection and diffraction phenomena at the edges of high-rise buildings. This behavior results from wave obstruction caused by the irregular geometries and the spatial distribution of urban structures.

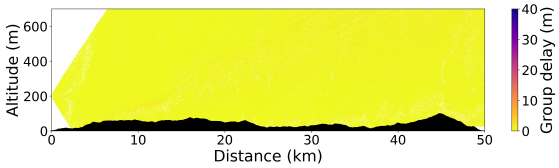
The extracted multipath components, shown in Fig. 15, illustrate the capability of the extraction method to disentangle multipath components, providing a understanding of wave behavior in environments with high obstacles.

TABLE IV
SOURCE AND SIMULATION PARAMETERS FOR ELECTROMAGNETIC WAVE PROPAGATION IN SIMULATION D.

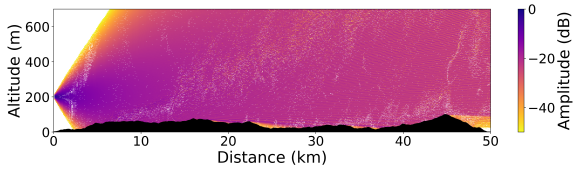
Parameters	Values	Parameters	Values
f	2 GHz	$z_{\max}$	1000 m
Polarization	TE	$z_{\max_cd}$	700 m
W_0	0.5 m	Δ_z	0.2 m
x_{w0}	-100 m	N_z	5000
y_s	0 m	$x_{\max}$	25 km
z_s	10 m	Δ_x	50 m
n_{f}	8	N_x	500
δ_{f}	1 MHz	n_{p}	4

V. CONCLUSION

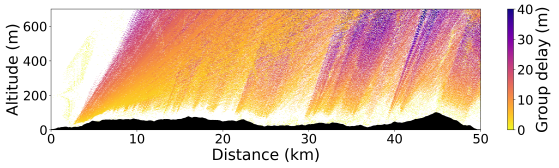
This study has investigated the impact of complex environments on CNS signal propagation, highlighting the need for accurate simulation and effective separation of multipath components. Two split-step simulation methods have been introduced for modeling electromagnetic wave propagation in the lower troposphere, effectively accounting for inhomogeneous atmospheric conditions, ground impedance, and irregular terrain. A multipath extraction method was subsequently proposed, relying on Prony's method to estimate the amplitude, phase, and delay of multipath components from simulation results at only a few frequencies. The accuracy and numerical performance of the method have been evaluated,



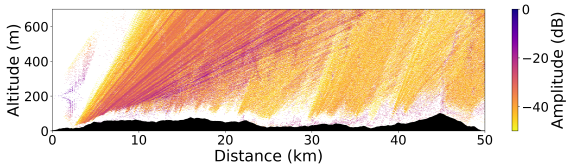
(a) Delay of the first extracted path component



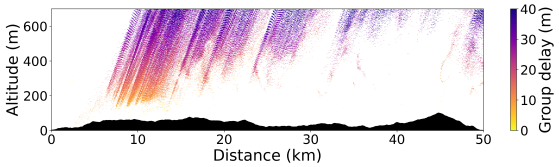
(b) Amplitude of the first extracted path component



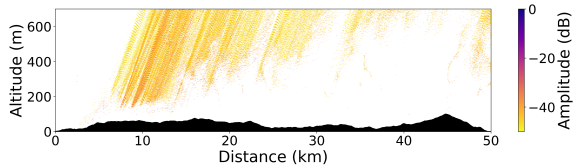
(c) Delay of the second extracted path component



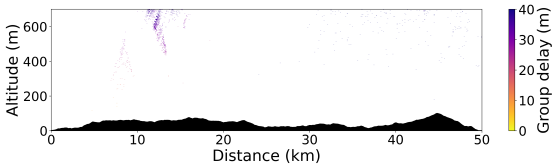
(d) Amplitude of the second extracted path component



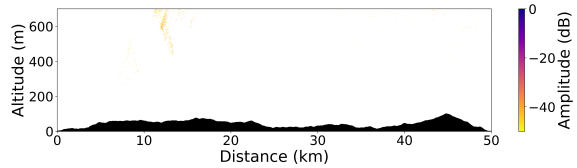
(e) Delay of the third extracted path component



(f) Amplitude of the third extracted path component



(g) Delay of the fourth extracted path component



(h) Amplitude of the fourth extracted path component

Fig. 10. Extracted multipath components in propagation over an irregular dielectric ground in a homogeneous atmosphere.

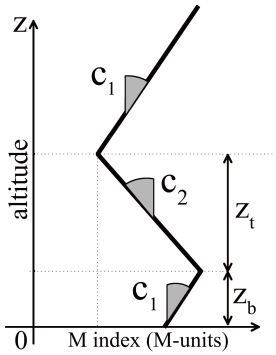


Fig. 11. Atmospheric modified refractivity profile.

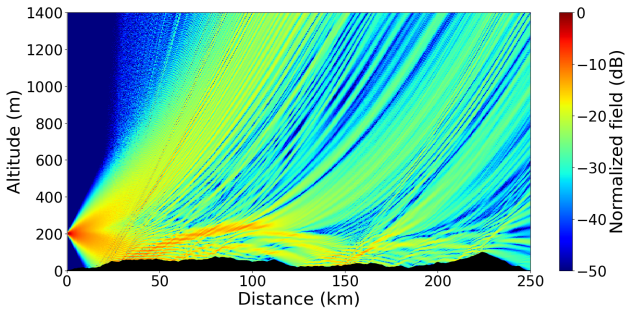
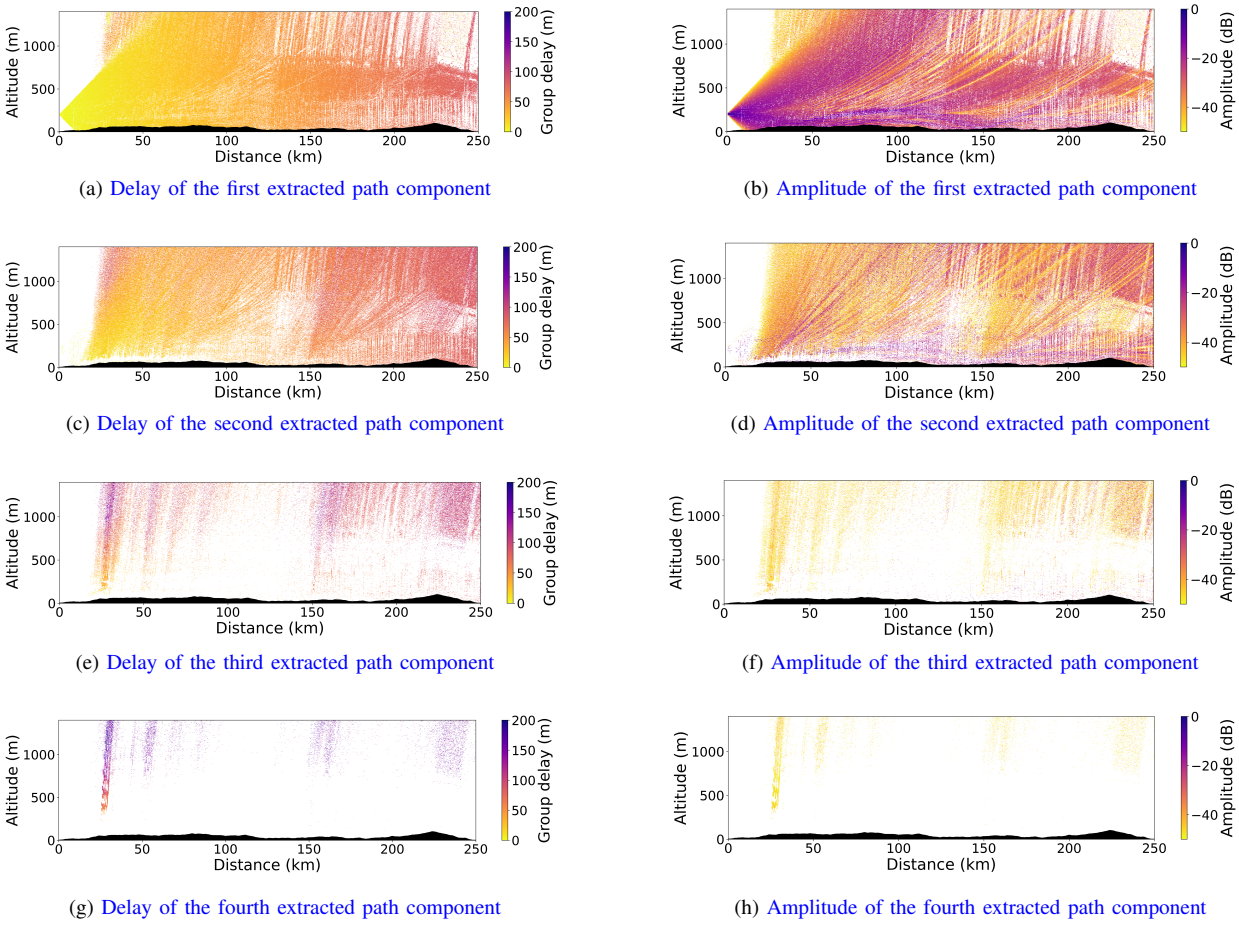


Fig. 12. Normalized electromagnetic wave field with SSW in propagation over an irregular dielectric ground within an atmospheric duct.

and a post-processing strategy has been suggested to enhance the interpretability of the results. To validate the method, simulations have been conducted under various environmental conditions. A comparison with the classical two-ray model with planar ground condition confirmed its accuracy. In that case, the extracted multipath components successfully reproduced the simulated channel, with a root mean square error below –50 dB across all spatial locations. However, a limitation is that the method performs less effectively when the delay spread between multipath components is small.

The proposed extraction approach is compatible with other frequency-domain electromagnetic simulation tools and can be extended to additional scenarios, such as urban canyons. It may also be adapted for use with two-way parabolic equation methods, particularly in configurations involving diffraction over vertical structures. Future research will involve broader comparisons with alternative extraction techniques in more complex environments, as well as validation through experimental measurements.

~~This study has examined the impact of complex~~



(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

(e) Delay of the third extracted path component

(f) Amplitude of the third extracted path component

(g) Delay of the fourth extracted path component

(h) Amplitude of the fourth extracted path component

Fig. 13. Extracted multipath components in propagation over an irregular dielectric ground within an atmospheric duct.

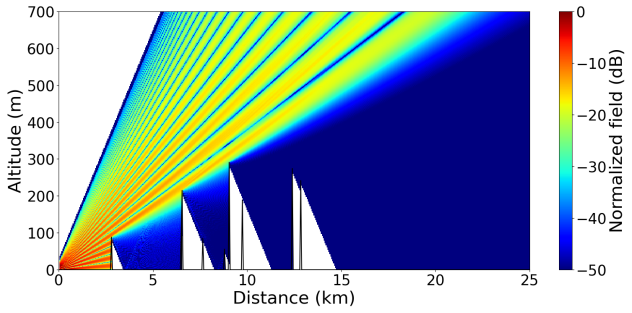


Fig. 14. Normalized electromagnetic wave field with SSW over vertical screens in a standard atmosphere (values below -50 dB are shown in white).

~~environments, which necessitate the simulation of CNS signals and the separation of multipath components. Two split-step simulation methods have been introduced for modeling electromagnetic wave propagation in the lower troposphere. These methods efficiently and accurately account for the effects of inhomogeneous atmospheric conditions, ground impedance, and irregular relief on wave propagation. Subsequently, a multipath extraction method has been proposed for separating multipath components based on the simulation results. The accuracy of the proposed method has been evaluated, and a post-processing approach has been suggested to enhance the interpretability of the results. Finally, numerical simulations have been conducted under various environmental conditions, demonstrating the accuracy and effectiveness of the proposed methods. The multipath extraction method can also be used with other computational electromagnetic tools in the frequency domain. This paves the way for additional scenarios, such as urban canyons.~~

ACKNOWLEDGEMENT

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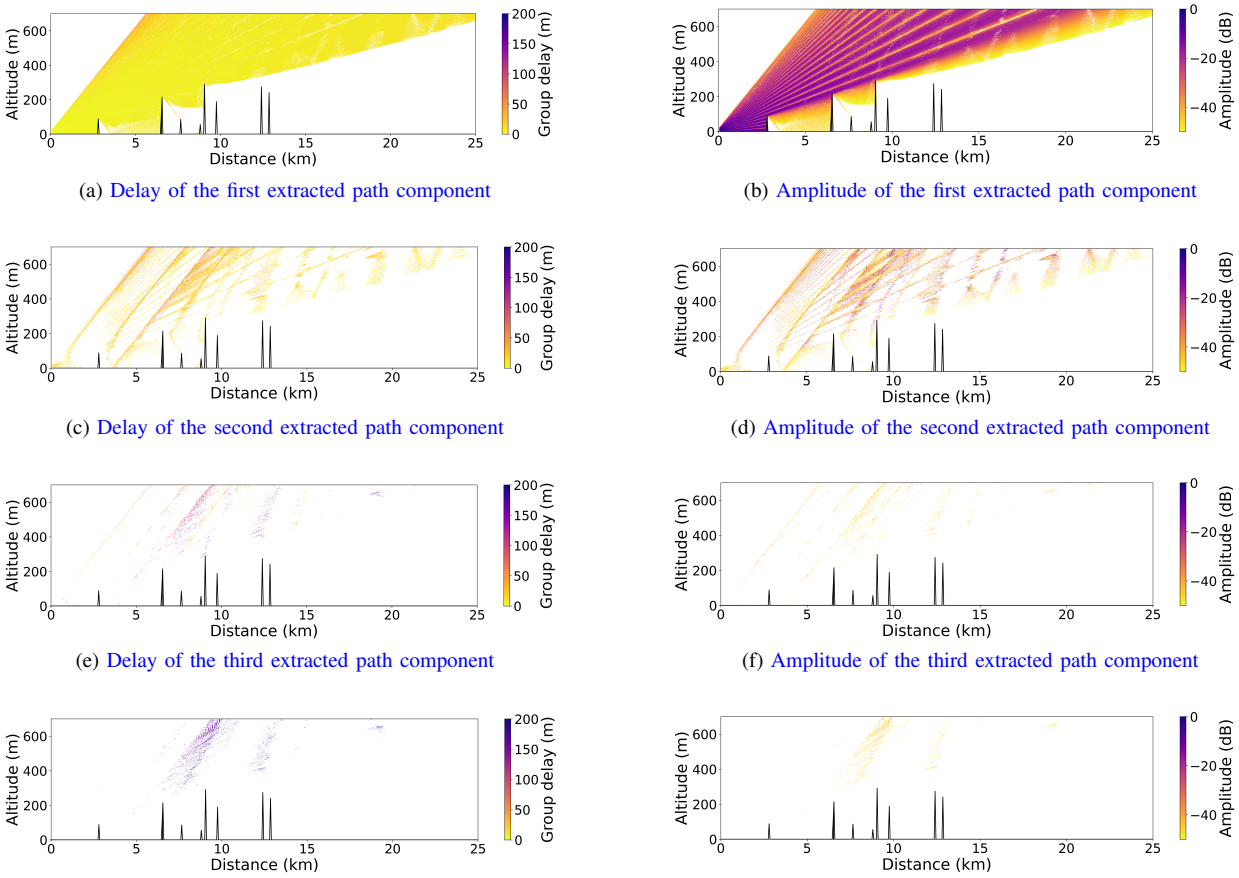

(a) Delay of the first extracted path component

(b) Amplitude of the first extracted path component

(c) Delay of the second extracted path component

(d) Amplitude of the second extracted path component

(e) Delay of the third extracted path component

(f) Amplitude of the third extracted path component

(g) Delay of the fourth extracted path component

(h) Amplitude of the fourth extracted path component

Fig. 15. Extracted multipath components in electromagnetic wave propagation over vertical screens in a standard atmosphere.

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